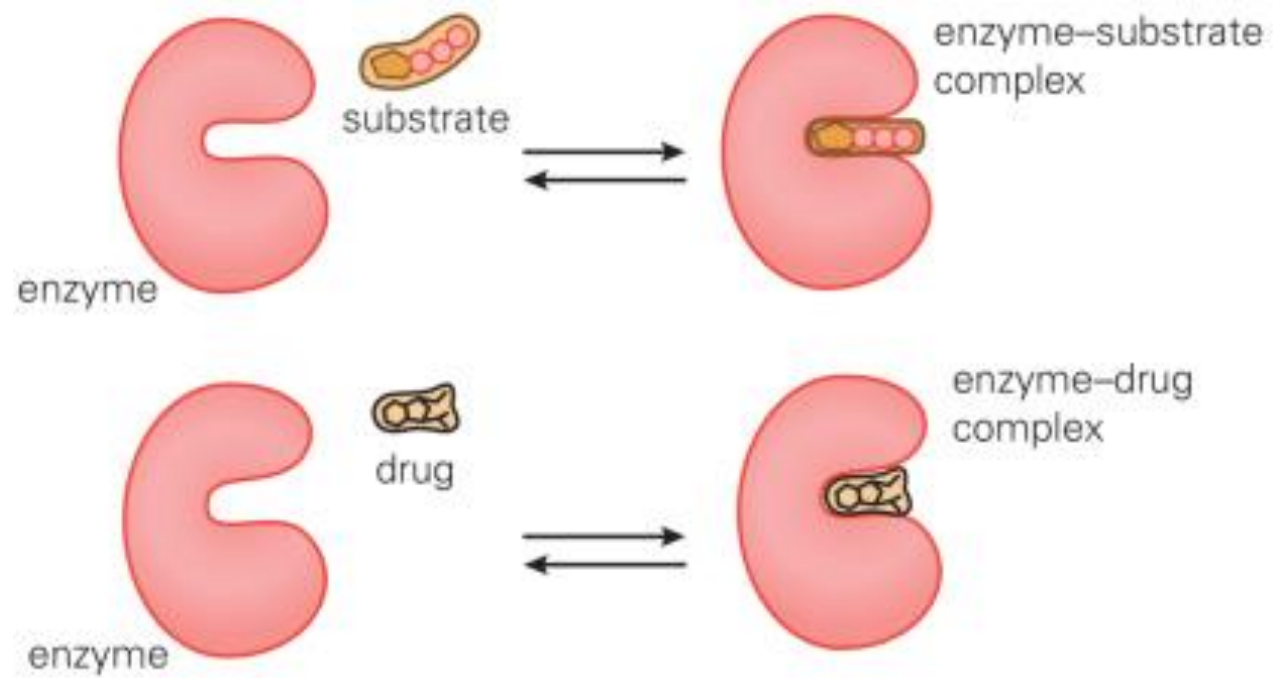


Protein-ligand complexes

Inhibitor/Drug	Disease/Condition	Enzyme Target
Acetazolamide	Glaucoma	Carbonic anhydrase
Acyclovir	Herpes	Viral DNA polymerase
Allopurinol	Gout	Xanthine oxidase
Argatroban	Coagulation	Thrombin
Aspirin, ibuprofen, DuP697	Inflammation, pain, fever	Prostaglandin synthase
β -Lactam antibiotics	Bacterial infections	D-Ala-D-Ala transpeptidase
Brequinar	Organ transplantation	Dihydroorotate dehydrogenase
Candoxatril	Hypertension, congestive heart failure	Atriopeptidase
Captopril	Hypertension	Angiotensin-converting enzyme
Clavulanate	Bacterial resistance	β -Lactamase
Cyclosporin	Organ transplantation	Cyclophilin/calcineurin
DuP450	AIDS	HIV protease
Enoximone	Congestive heart failure ischemia	cAMP phosphodiesterase
Finasteride	Benign prostate hyperplasia	Testosterone-5- α -reductase
FK-506	Organ transplantation, autoimmune disease	FK-506 binding protein
Fluorouracyl	Cancer	Thymidilate synthase
3-Fluorovinylglycine	Bacterial infection	Alanine racemase
(2-Furyl)-acryloyl-Gly- Phe-Phe	Lung elastin degradation in cystic fibrosis	<i>Pseudomonas</i> elastase
ICI-200,808	Emphysema	Neutrophil elastase
Lovastatin	High cholesterol	HMG CoA reductase

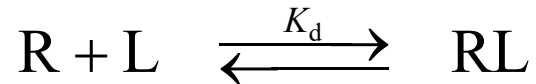


Strength of non-covalent interactions

Interaction	Strength (kJ/mol)
Dispersive (vdW)	0.2-2.0
Dipole-dipole	0.5-15
Ion-dipole	5-60
Hydrogen bond	4-40
Electrostatic (Ion-ion)	40-400
Hydrophobic*	~ 0.1
Covalent	200-800

* - kJ/mol/Å² (per non-polar surface area)

Quantifying binding



$$K_d = \frac{[R][L]}{[RL]} \quad (\text{Equilibrium dissociation constant})$$

K_d is in units of concentration, like K_m (Michaelis cte.) and K_i (inhibition cte.)

$$\Delta G_{\text{binding}} = -RT \ln K_{\text{eq}} = RT \ln K_d$$

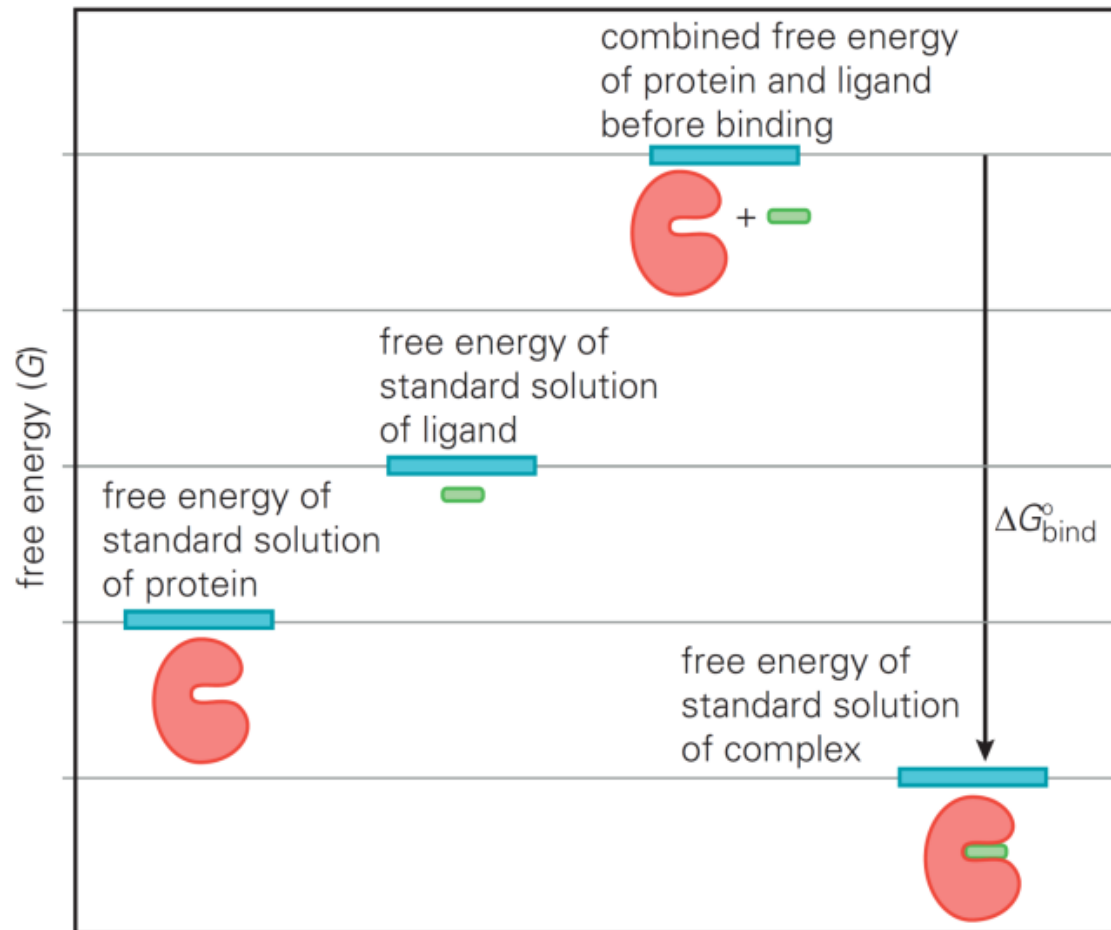
K_d and $\Delta G_{\text{binding}}$

K_d (M)	$\Delta G_{\text{binding}}$ (kJ)
1	0
10^{-1}	-5.9
10^{-2}	-11.9
10^{-3} (mM)	-17.8
10^{-4}	-23.7
10^{-5}	-29.7
10^{-6} (μ M)	-35.6
10^{-7}	-41.5
10^{-8}	-47.5
10^{-9} (nM)	-53.4
10^{-10}	-59.4
10^{-11}	-65.3
10^{-12} (pM)	-71.2



strong binding

$\Delta G_{\text{binding}}$ depends on the environment

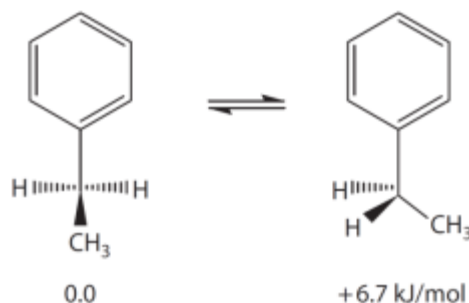


$\Delta G_{\text{binding}}$ is small

Increasing K_d by a factor of 10 results in a change ($\Delta\Delta G$) of only 5.7 kJ/mol:

$$\Delta\Delta G_{\text{binding}} = RT \ln K'_d - RT \ln K_d = RT \ln \frac{K'_d}{K_d} = RT \ln 10 = 5.7 \text{ kJ/mol}$$

Even small conformational changes can produce energy changes of this magnitude:

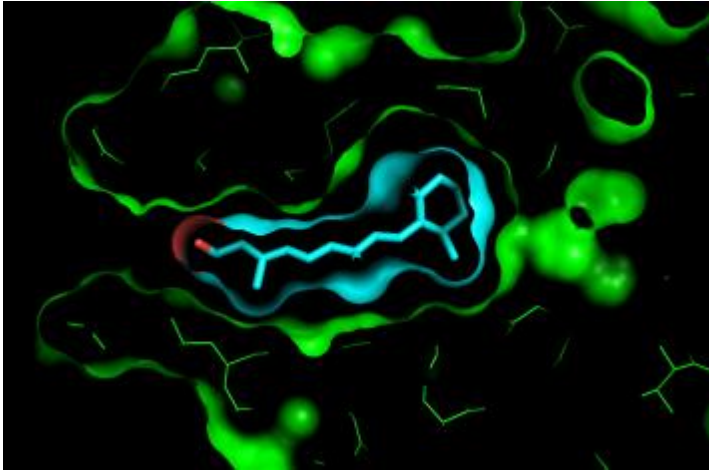


Modest changes in conformation can produce large variations in affinity!

Table 12.1 Typical strengths of different kinds of interactions.

Type of interaction	K_D (molar)	$\Delta G_{\text{bind}}^{\circ}$ (at 300 K) (kJ•mol ⁻¹)
Enzyme–ATP	$\sim 1 \times 10^{-3}$ to $\sim 1 \times 10^{-6}$ (millimolar to micromolar)	–17 to –35
Signaling protein binding to a target	$\sim 1 \times 10^{-6}$ (micromolar)	–35
Sequence-specific recognition of DNA by a transcription factor	$\sim 1 \times 10^{-9}$ (nanomolar)	–52
Small molecule inhibitors of proteins (drugs)	$\sim 1 \times 10^{-9}$ to $\sim 1 \times 10^{-12}$ (nanomolar to picomolar)	–52 to –69
Biotin binding to avidin protein (one of the strongest known noncovalent interactions)	$\sim 1 \times 10^{-15}$ (femtomolar)	–86

Lock-and-key model



Vitamin A bound to its
transporter protein

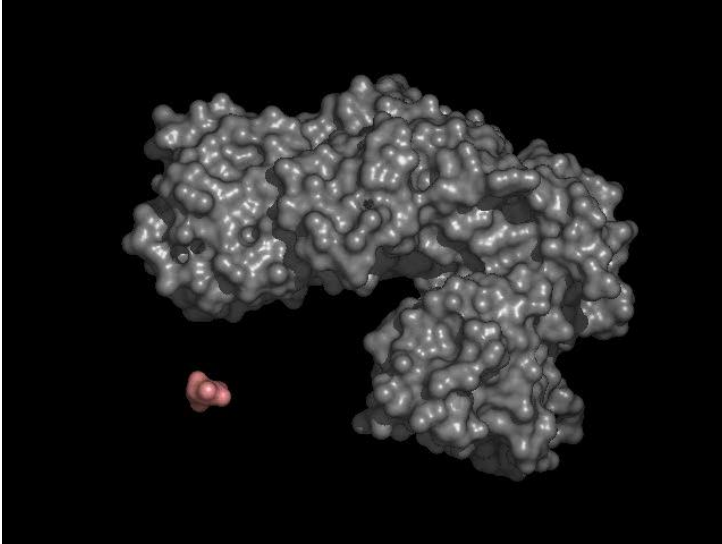


Hermann Emil Fischer
(1852-1919)

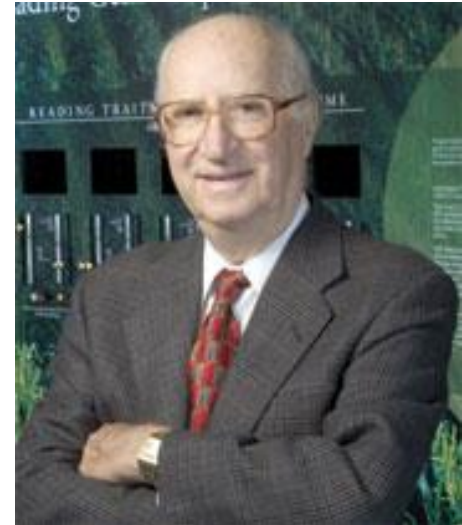
*The limited effect of enzymes on the glucosides can also be explained by the assumption that a chemical process can be initiated only by those [enzymes] that have a similar geometric construction that approximates that of the molecule [substrates]. To use a picture, I want to say that enzymes and glucosides must fit together like a **lock and key** to be able to exert a chemical effect upon one another. This idea has gained plausibility and value for stereochemistry research after the phenomena was transferred from the biological to the chemical field.*

Reports of the German Chemical Society 1894

Induced fit



Binding of glucose by hexokinase



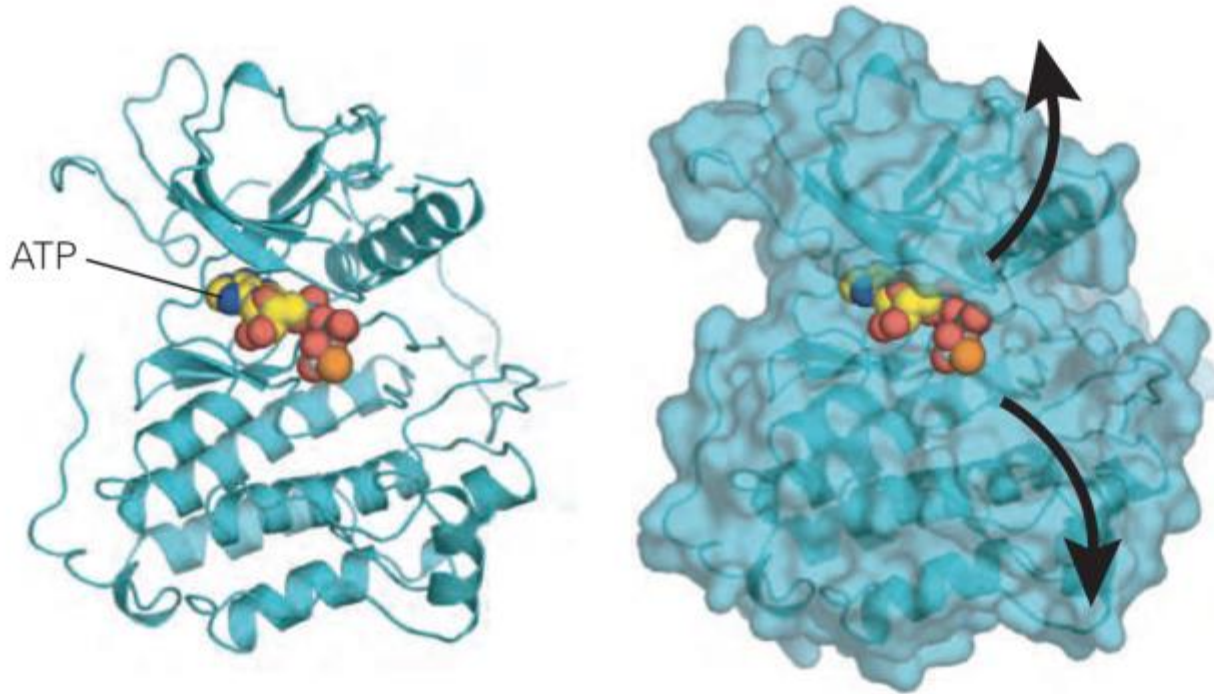
Daniel Koshland, Jr.
(1920-2007)

“My first inkling that the Fischer key-and-lock model needed modifications really arose from my consideration of the role of water in biological reactions. I was preparing a lecture for a scientific meeting and decided to consider why some proteins were kinases and others ATPases. The more I thought about the protein, the more astonishing it seemed that water could be prevented from reaching the active site. ”

D.E. Koshland, Jr.

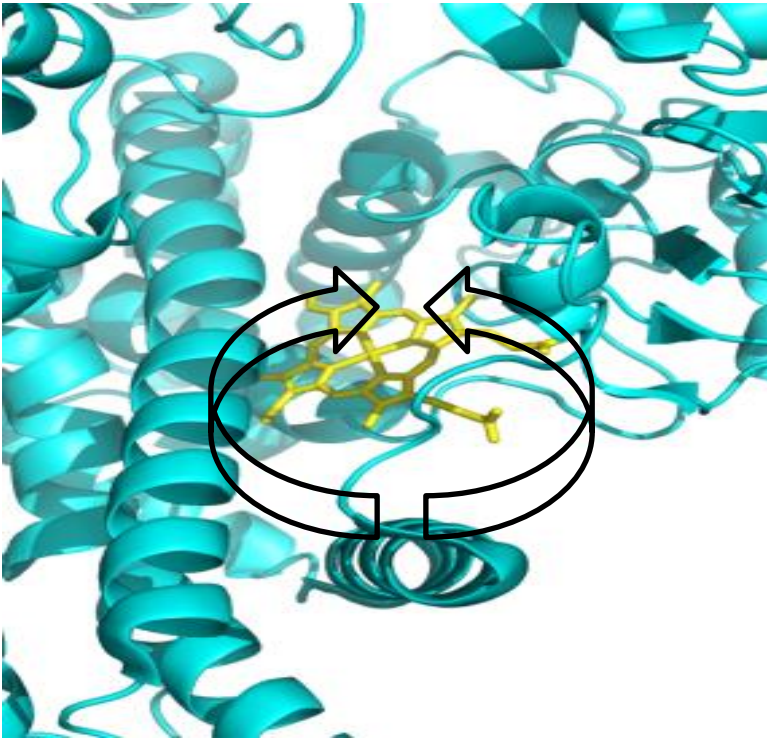
D. E. Koshland. Jr., *PNAS* **1958**. 44, 98

Induced fit is necessary

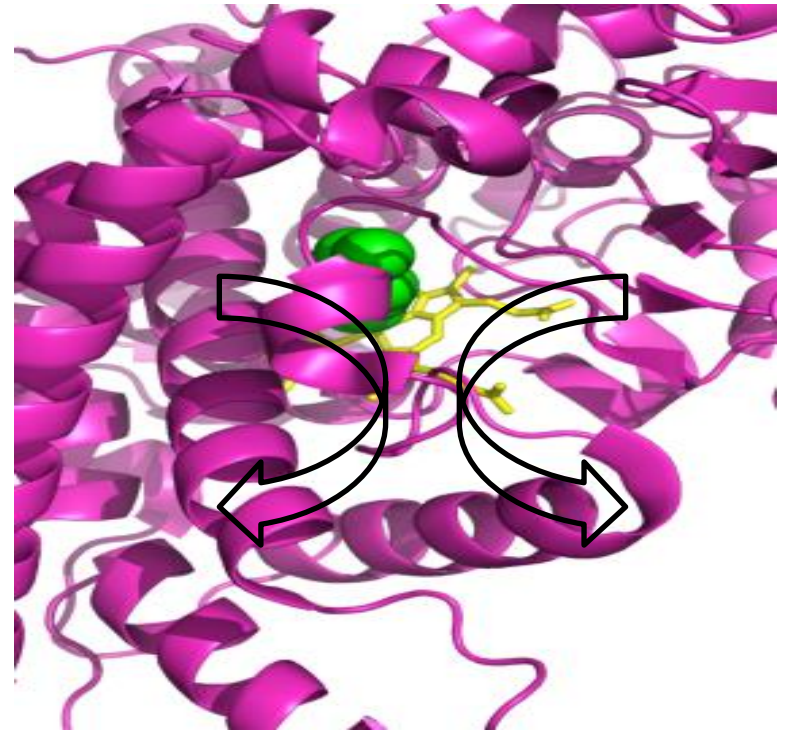


ATP bound to the EGF receptor kinase domain

Two conformations of CYP2B4



**Open
conformation**



**Closed
conformation**

Only the left conformation allows binding of the ligands!

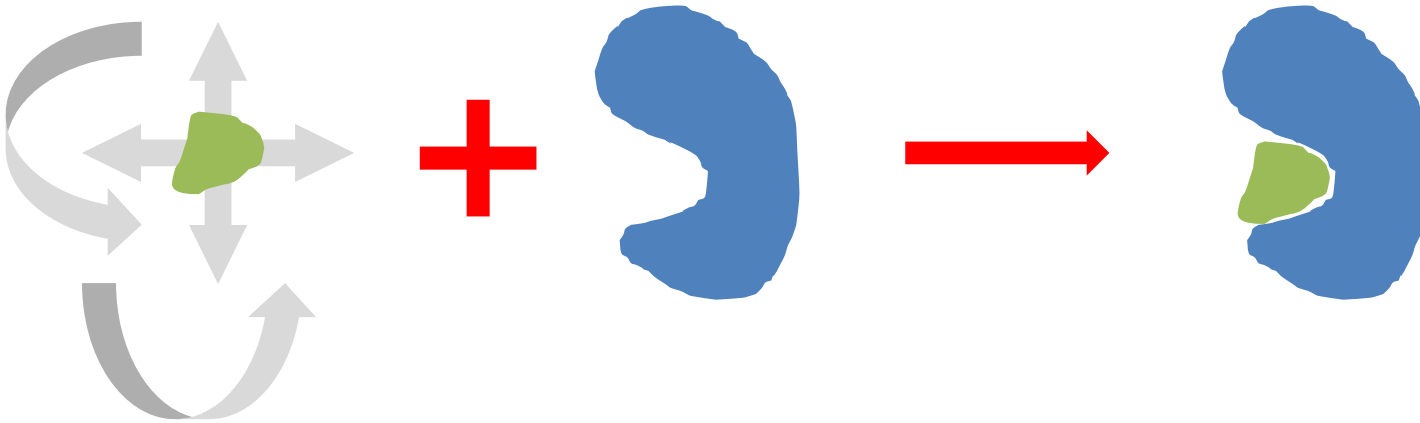
The thermodynamics of binding

The binding free energy can be partitioned in different components:

$$\Delta G_{\text{binding}} = \Delta G_{\text{transl+rot}} + \Delta G_{\text{conf}} + \Delta G_{\text{polar}} + \Delta G_{\text{hydrophob}} + \Delta G_{\text{vdW}}$$

- $\Delta G_{\text{transl+rot}}$ → restrictions in translational movements
- ΔG_{conf} → change in number of available conformational states
- ΔG_{polar} → polar interactions (including H-bonding)
- $\Delta G_{\text{hydrophob}}$ → binding energy due the hydrophobic Effect
- ΔG_{vdW} → binding energy due to van der Waals interactions

$$\Delta G_{\text{trans+rot}}$$



Loss of translational and rotational freedom upon binding, generally unfavorable, because:

$$\Delta S < 0 \Rightarrow -T \Delta S > 0,$$

resulting in a positive term (destabilizing) in the Gibbs free energy:

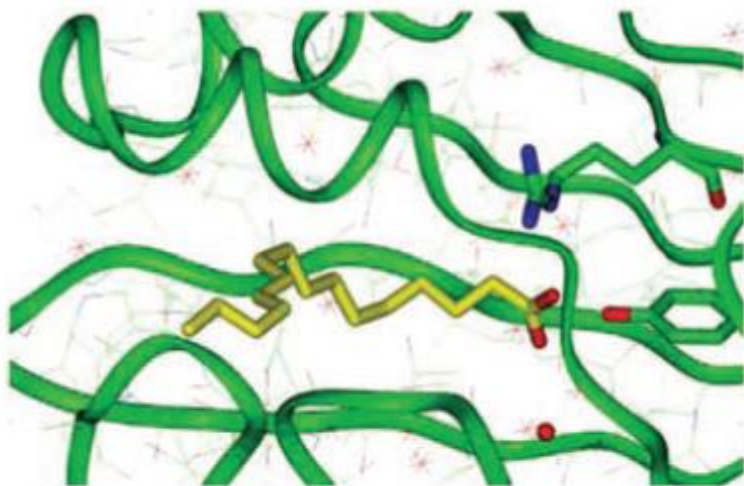
$$\Delta G = \Delta H - T \Delta S$$

$$\Delta G_{\text{conf}}$$



Loss of conformational degrees of freedom upon binding . Unfavorable due to $\Delta S < 0$.

$$\Delta G_{\text{conf}}$$



(a)



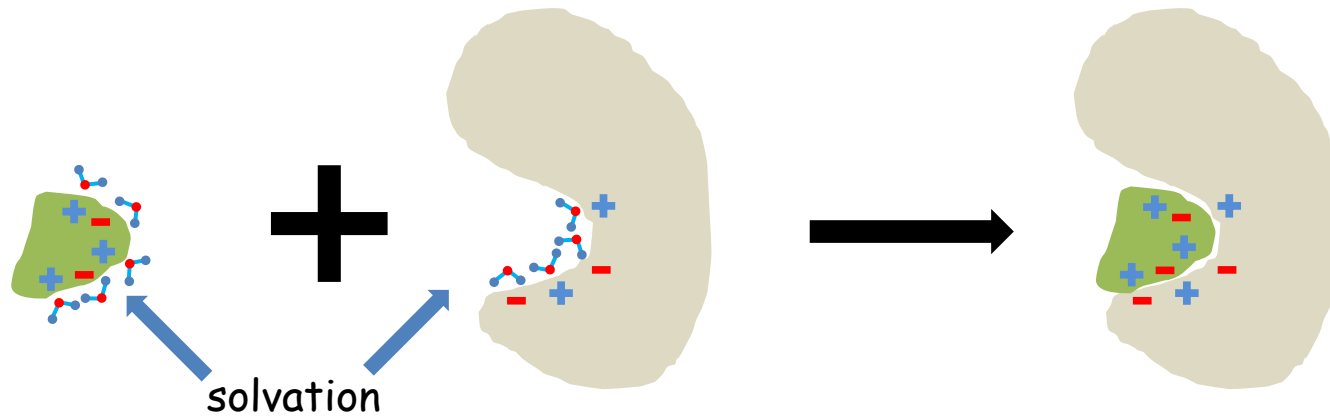
(b)



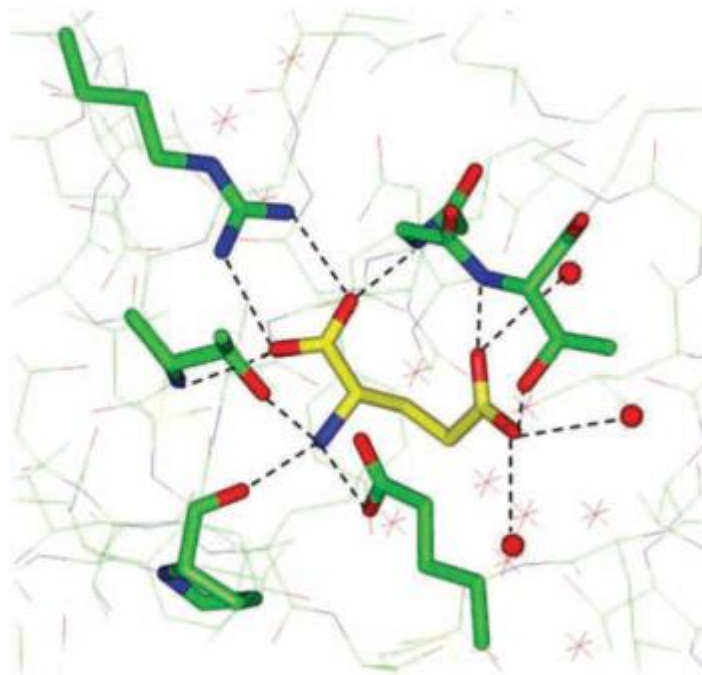
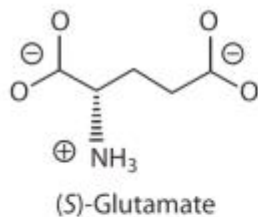
Bioactive conformation
+10.5 kJ/mol

The ligand may bind in an energetically unfavorable conformation, further adding a positive penalty to ΔG_{conf} !

$$\Delta G_{\text{polar}}$$

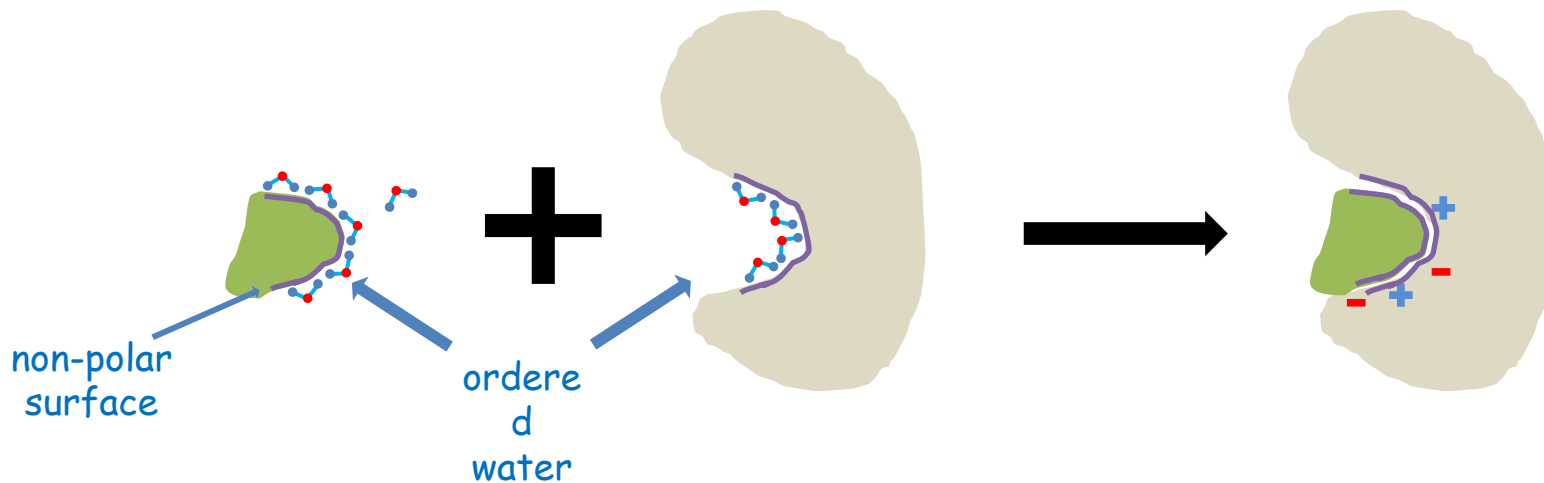


Electrostatic and polar forces are strong and long range, but their contribution to binding is small due the strong solvation interactions in the unbound state.



The net stabilization of H-bonds is small, but their directionality is a powerful tool for molecular recognition (and unsatisfied hydrogen bonds have a large penalty).

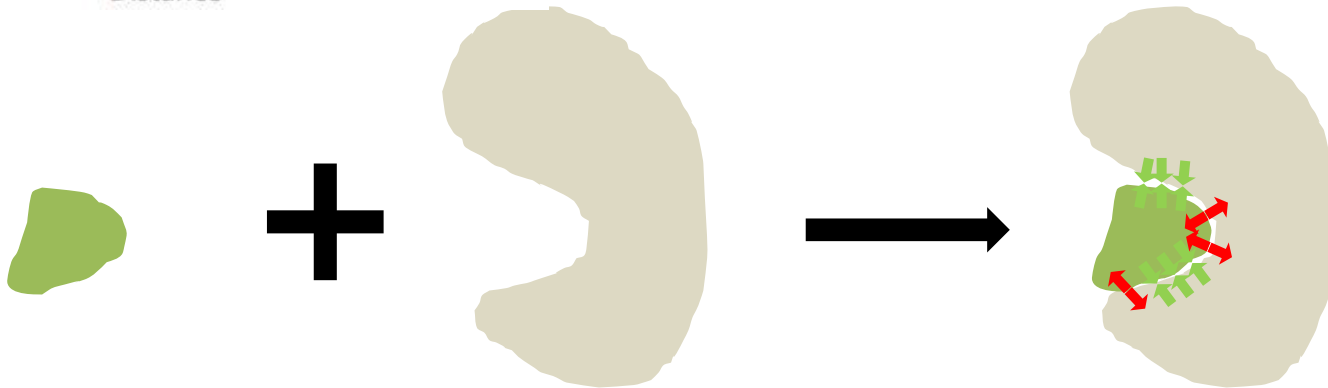
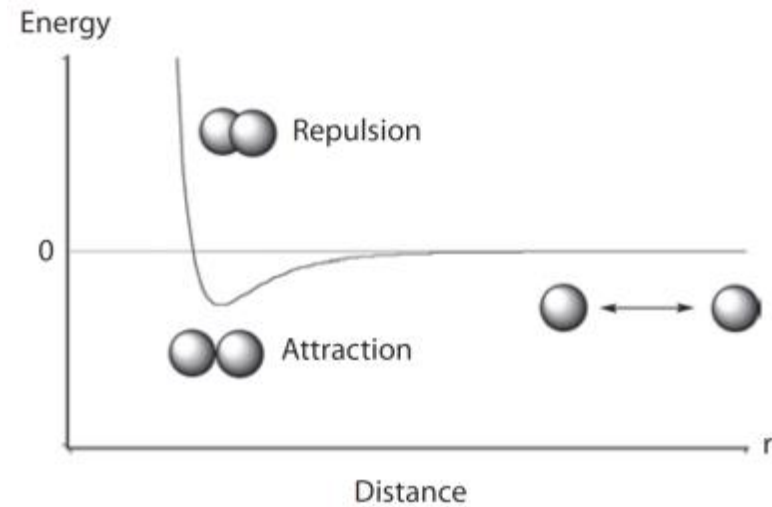
$$\Delta G_{\text{hydrophob}}$$



The burial of non polar surface releases the ordered water layers, providing an increase in entropy that is a major force for binding.

The magnitude of the hydrophobic effect is related to the buried area upon binding. Based on experiments, the magnitude range is **0.1-0.3 kJ/Å²**.

$$\Delta G_{\text{vdW}}$$



VDW forces have repulsive and attractive terms that describe weak (~ 0.2 kJ/M), but very numerous atom-atom interactions.

Affinity *is not* specificity

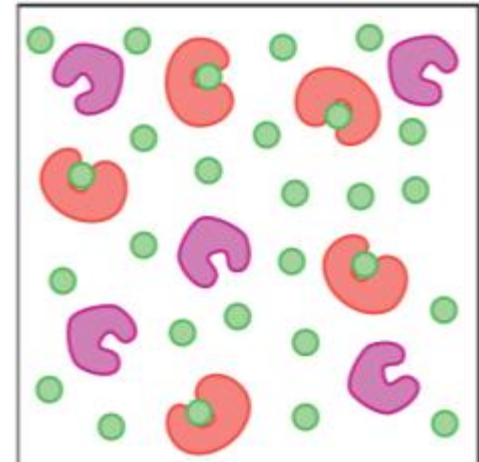
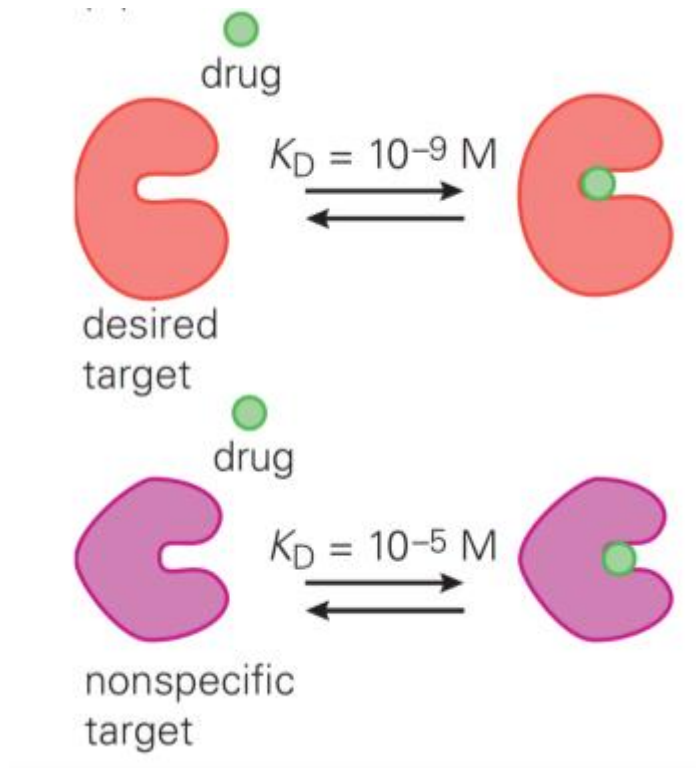
- **Affinity** – measure of interaction strength between binding partners
- **Specificity** – ability of a molecular species to discriminate between multiple partners

Specificity is always relative to the repertoire of available partners

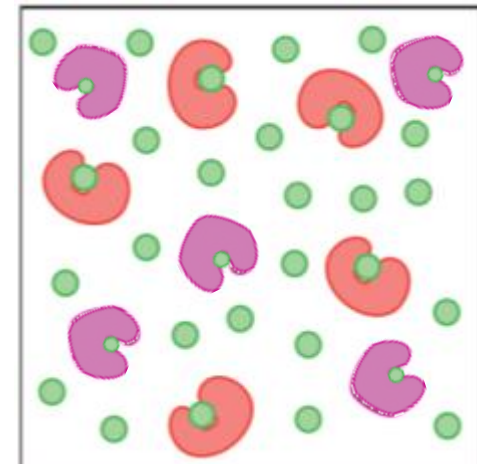
Davenport's rule:

ligand specificity is inversely proportional to knowledge about the ligand 😊

Specificity



Drug concentration = $0.1 \mu\text{M}$



Drug concentration = 0.1 M

Specificity depends on concentration !!

Affinity and specificity may have a biological context

Table 1.2. Affinity of Some Stereoisomers for Chymotrypsin * [44]

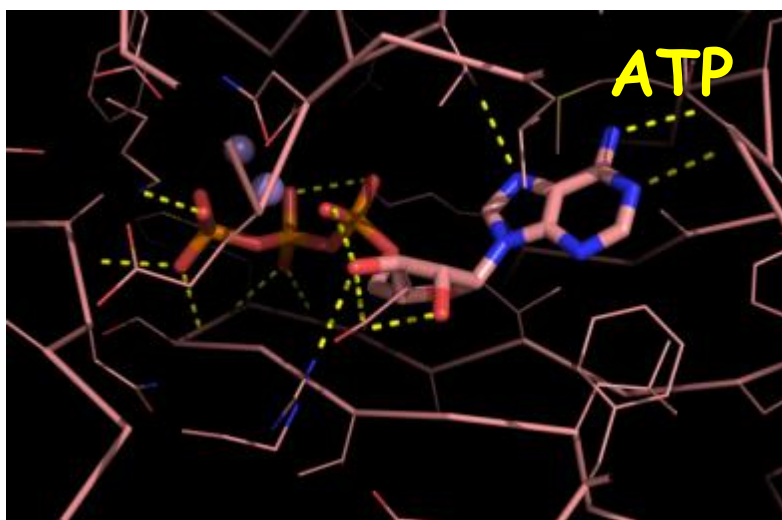
Compound	K_M , mM	K_I , mM	Compound	K_M , mM	K_I , mM
Acetyl-L-tryptophanamide	5.5		Nicotinyl-L- tyrosinamide	15.0	
Acetyl-D-tryptophanamide		2.7	Nicotinyl-D- tyrosinamide		6.2
Acetyl-L-tyrosinamide	30.5		Acetyl-L- tryptophanamide		17.5
Acetyl-D-tyrosinamide		12.0	Acetyl-D- tryptophanamide		4.8
Nicotinyl-L- tryptophanamide	2.7		L- tryptophanamide		6.3
Nicotinyl-D- tryptophanamide		1.4	D- tryptophanamide		3.2

* For chymotrypsin values of K_s (K_I) и K_M are close [45]

The naturally occurring L-amino acids bind with less affinity ($K_M > K_I$) than the artificial D-amino acids. Specificity was not positively selected because the D-amino acids are not present in the biological context.

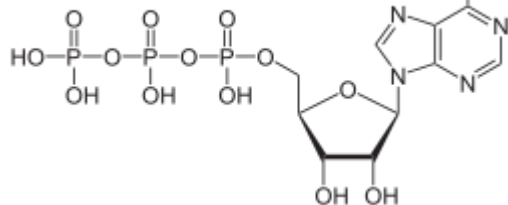
Something to have in mind when studying

Hydrophobicity dominates affinity

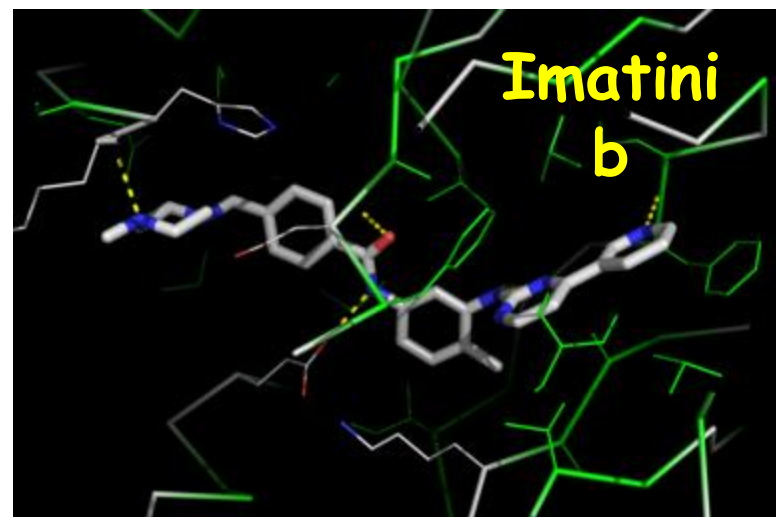


$K_D = 10 \mu\text{M}$

More H-bonds

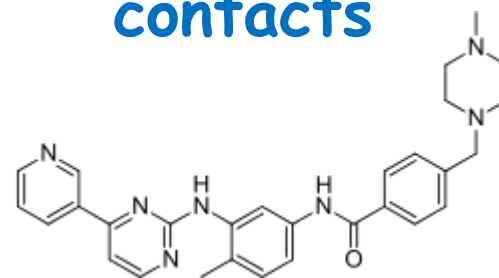


ATP



$K_D = 10 \text{ nM}$

More hydrophobic contacts

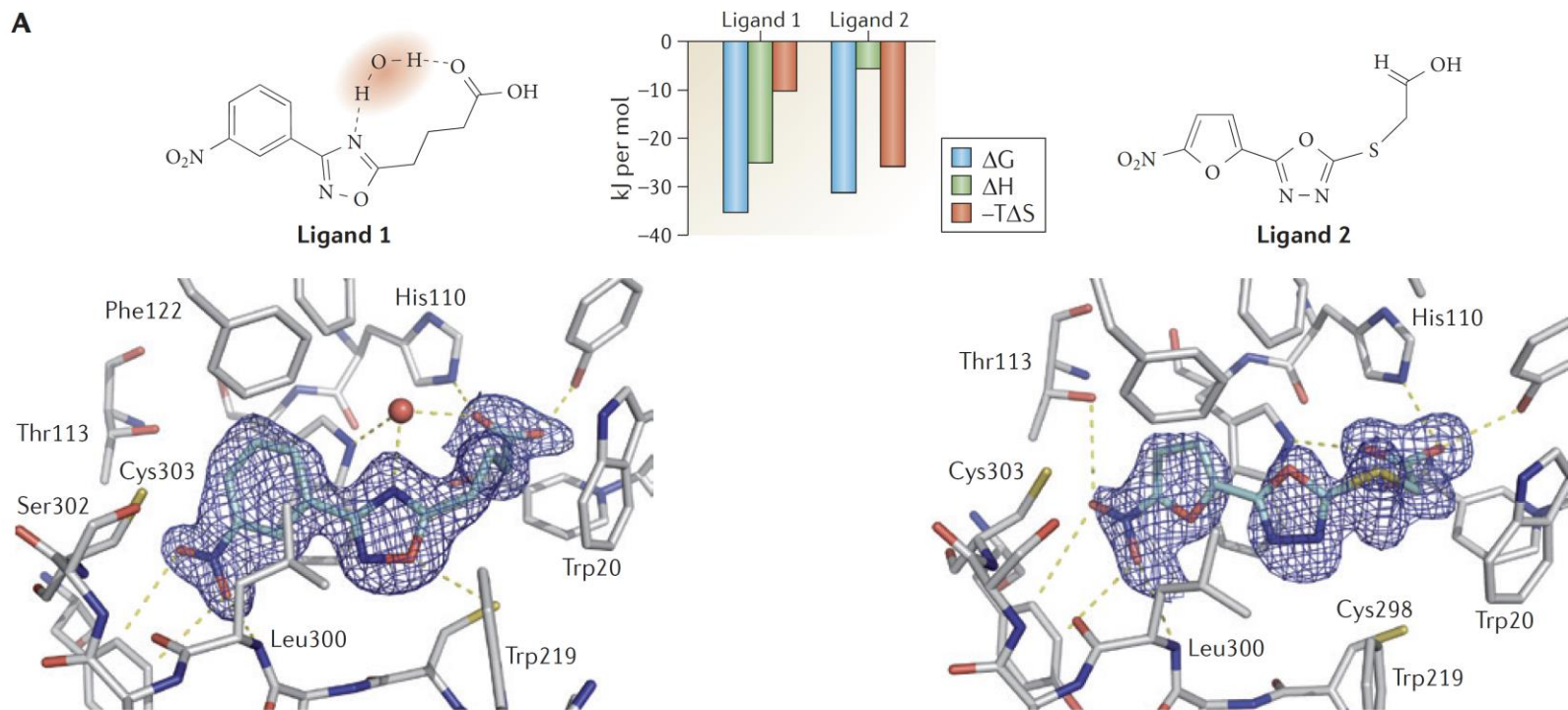


Imatinib

The K_D for ATP is 1000x higher than that of the drug Imatinib.

Water molecules at the binding site can play an important role

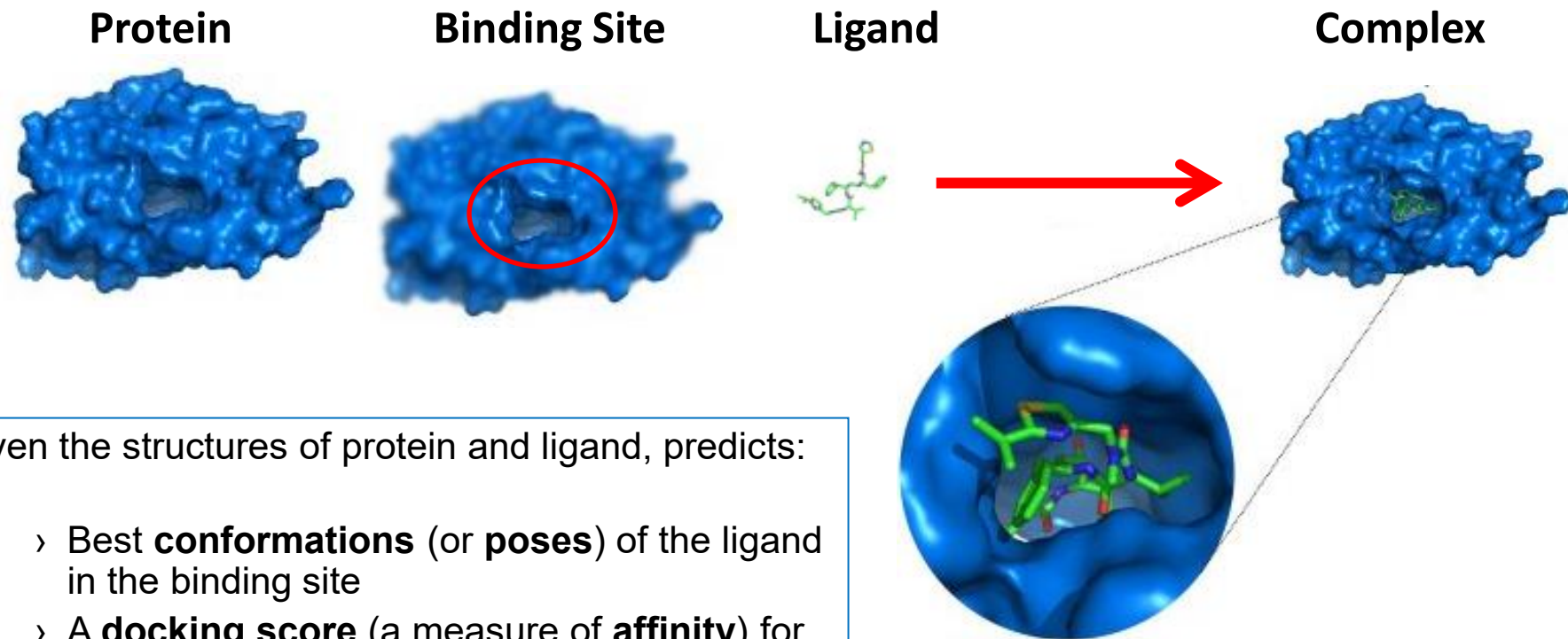
A



Protein-ligand docking

What is protein-ligand docking ?

- Structure-based method (SBDD)
 - “structure” means “using the structure of protein and ligand”
- Computational techniques that mimic (approximately) the process of protein-ligand binding



Given the structures of protein and ligand, predicts:

- › Best **conformations** (or **poses**) of the ligand in the binding site
- › A **docking score** (a measure of **affinity**) for each pose of the ligand on the binding site

Why docking matters for drug design

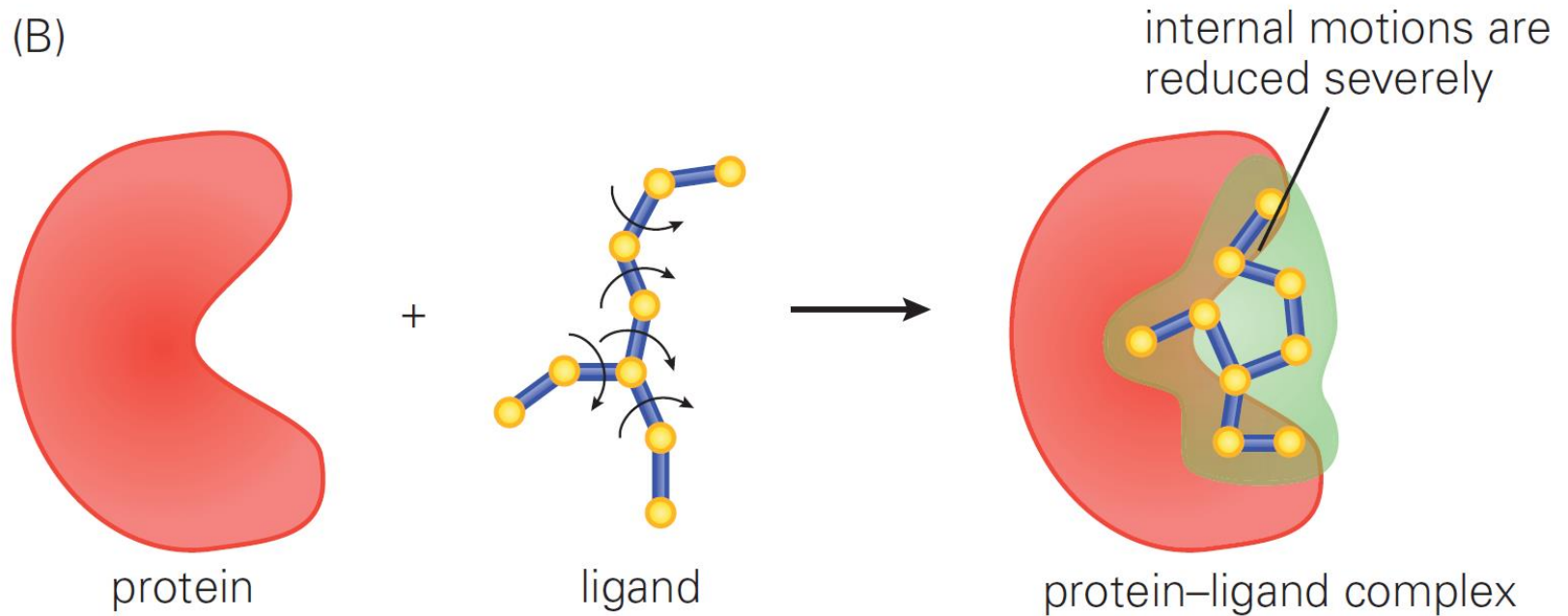
- **Rational Drug Design:** From Serendipity to Strategy
 - Moves drug discovery beyond trial-and-error. Docking allows researchers to *design* molecules that fit a specific protein target (like a key in a lock), enabling a more rational, efficient, and targeted approach to creating new therapeutics.
- **Predicting Binding Affinity:** Identifying Promising Candidates Early
 - By scoring how tightly a ligand binds to its target, docking helps prioritize the most potent molecules from millions of virtual compounds. This drastically reduces the time and cost spent on synthesizing and testing weak candidates in the lab.
- **Elucidating Mechanisms of Action:** Understanding How Drugs Work
 - Docking provides a 3D visual model of *how* a drug interacts with its protein target at the atomic level. This reveals the critical interactions (hydrogen bonds, hydrophobic contacts) responsible for the drug's effect, guiding the optimization of its properties.
- **Virtual High-Throughput Screening:** Sifting Through Vast Chemical Libraries
 - It is computationally feasible to "dock" millions of small molecules from virtual libraries into a protein's active site. This acts as a powerful computational filter, identifying a handful of top "hits" for experimental testing, accelerating the initial discovery phase.
- **Structure-Based Lead Optimization:** Fine-Tuning Drug Properties
 - For a promising but imperfect lead compound, docking shows exactly which parts of the molecule can be modified to improve binding strength, selectivity (to reduce side effects), and other drug-like properties, guiding medicinal chemists in their synthesis efforts.

Rigid versus flexible docking

- **The Core Difference: Flexibility of Molecules**
 - Rigid Docking: Treats both the protein receptor and the ligand as static, unchanging shapes.
 - Flexible Docking: Allows the ligand to change its shape (conformation) to fit the protein's binding site.
- **The "Lock and Key" vs. "Induced Fit" Analogy**
 - Rigid Docking: Follows the "Lock and Key" model, assuming a perfect, pre-formed fit.
 - Flexible Docking: Follows the "Induced Fit" model, where both the ligand and the binding site can adjust to achieve optimal complementarity.
- **Computational Cost vs. Accuracy**
 - Rigid Docking: Is computationally fast and simple, allowing for the rapid screening of millions of compounds. However, it can lack accuracy.
 - Flexible Docking: Is computationally expensive and slower because it explores many more possible conformations. The benefit is a more physically accurate and reliable result.
- **Ideal Use Cases**
 - Rigid Docking: Best for initial, high-throughput **virtual screening** to quickly filter large compound libraries for potential "hits."
 - Flexible Docking: Used for later-stage **lead optimization** to understand precise binding modes and improve the potency of the most promising candidates.

Rigid receptor, flexible ligand

(B)



Docking

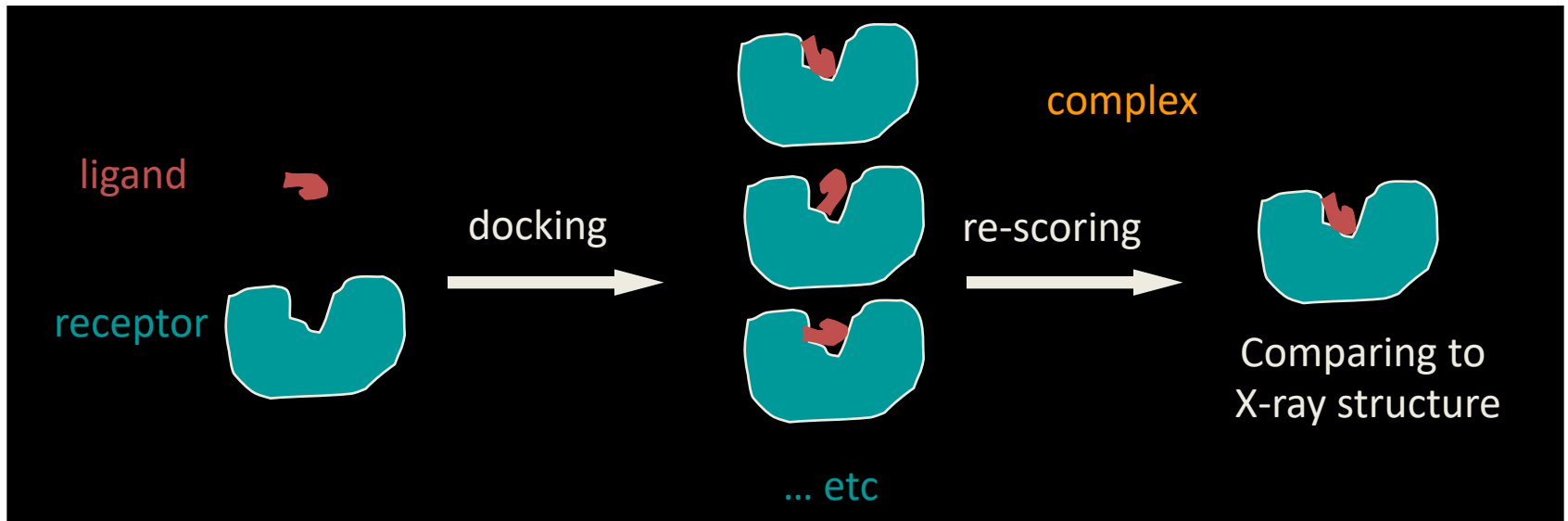
- **Problema 1:** encontrar a conformação (ou conformações) de mais baixa energia para associação de duas moléculas
- **Problema 2:** calcular diferenças de energia livre entre ligandos de uma mesma proteína (semi-quantitativo)
- **Problema 3:** calcular energias livres *absolutas* de ligação de ligandos a proteínas (desenvolvimento de métodos)

Ordem de dificuldade: **1,2,3** . Para o problema 1, o mais fácil, conseguem-se bons resultados em muitos casos. Para o problema 3 ainda estamos longe de ter uma solução rigorosa e geral.

Docking e scoring

The discovery of optimal receptor-ligand conformations involves two distinct steps:

- **Conformational search (docking):** finding the solutions that yield the lowest score. This is a minimization problem.
- **Conformation evaluation (scoring):** ranking the found conformations in descending order of energy, calculated by the same or a new, more specialized scoring function (re-scoring)



Docking software

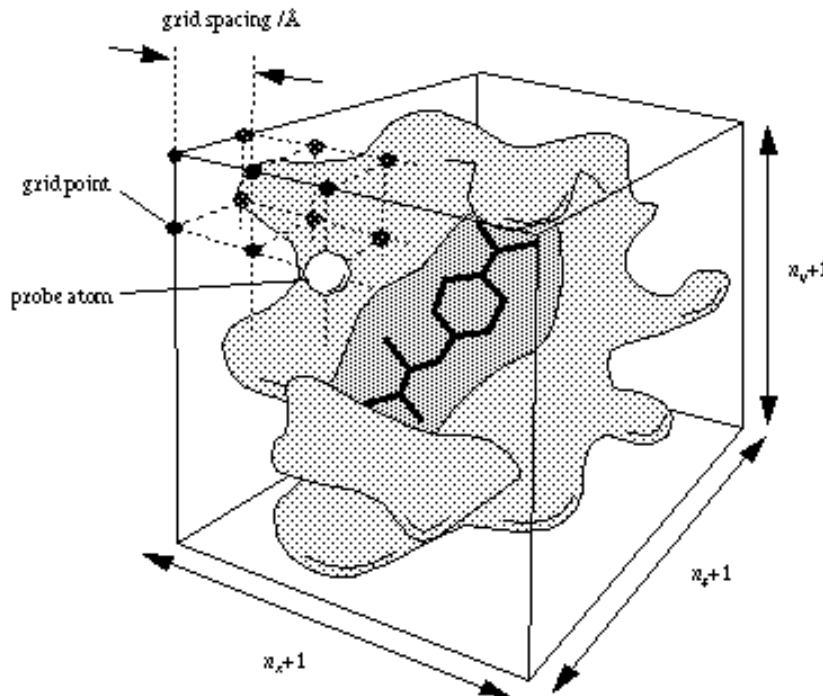
Program Name	License Type	Specific Notes & Strengths
AutoDock Vina	Free & Open-Source	The most widely used docking program. Excellent balance of speed and accuracy. Simple to use.
AutoDock4	Free & Open-Source	Predecessor to Vina; more established but slower. Known for high accuracy with proper setup.
LeDock	Free for Academia	Very fast and user-friendly with a simple configuration file. Great for benchmarking.
rDock	Free & Open-Source	Excellent for virtual screening workflows. Strong support for solvation and constraints.
CB-Dock	Free & Open-Source	Web server available. Automatically detects binding sites, making it ideal for beginners.
Schrödinger Glide	Commercial	Industry gold standard for accuracy. Uses a hierarchical filtering approach for high-speed precision.
GOLD	Commercial	Uses a genetic algorithm . Renowned for reliability and handling protein flexibility.
MOE	Commercial	An all-in-one software suite. Excellent for structure-based design and QSAR in a user-friendly interface.
ICM-Pro	Commercial	Excellent handling of full ligand and receptor flexibility using internal coordinate mechanics.
HADDOCK	Free for	Specialized Tool: Driven by experimental data (NMR, cryo-EM). Best for protein-

Autodock 4 Energy Function

$$\Delta G_{\text{bind}} = \Delta G_{\text{vdW}} + \Delta G_{\text{H-bond}} + \Delta G_{\text{elec}} + \Delta G_{\text{desolv}} + \Delta G_{\text{tors}}$$

$$\begin{aligned} \Delta G_{\text{bind}} = & \sum_{i=1}^{N_{\text{lig}}} \sum_{j=1}^{N_{\text{rec}}} \left[\frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^6} \right] \text{ van der Waals} \\ & + E(t) \left(\frac{C_{ij}}{r_{ij}^{12}} - \frac{D_{ij}}{r_{ij}^{10}} \right) \text{ Hydrogen bonds} \\ & + \frac{q_i q_j}{\varepsilon(r_{ij}) r_{ij}} \text{ Electrostatic} \\ & + \left[\frac{S_i S_j (V_i + V_j)}{2} \exp \left(-\frac{r_{ij}^2}{2\sigma^2} \right) \right] \text{ Desolvation} \\ & + N_{\text{tors}} \cdot \Delta G_{\text{tors}}^{\text{const}} \text{ Torsion count} \end{aligned}$$

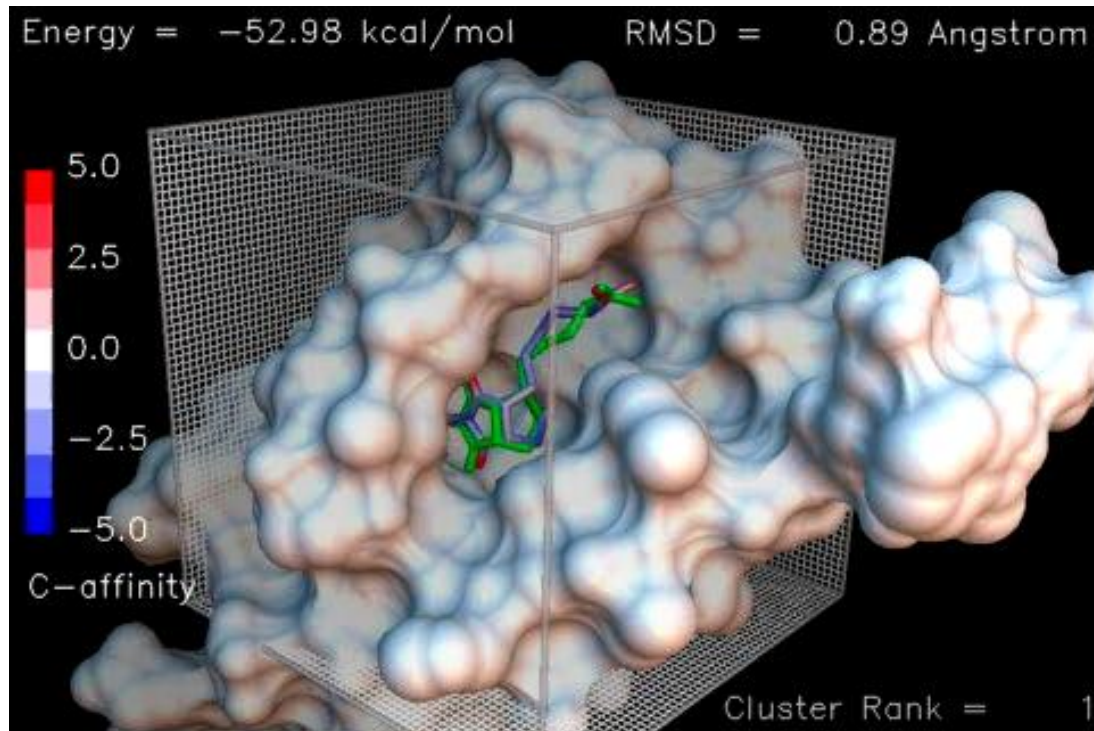
Avaliação da energia de interacção numa grelha



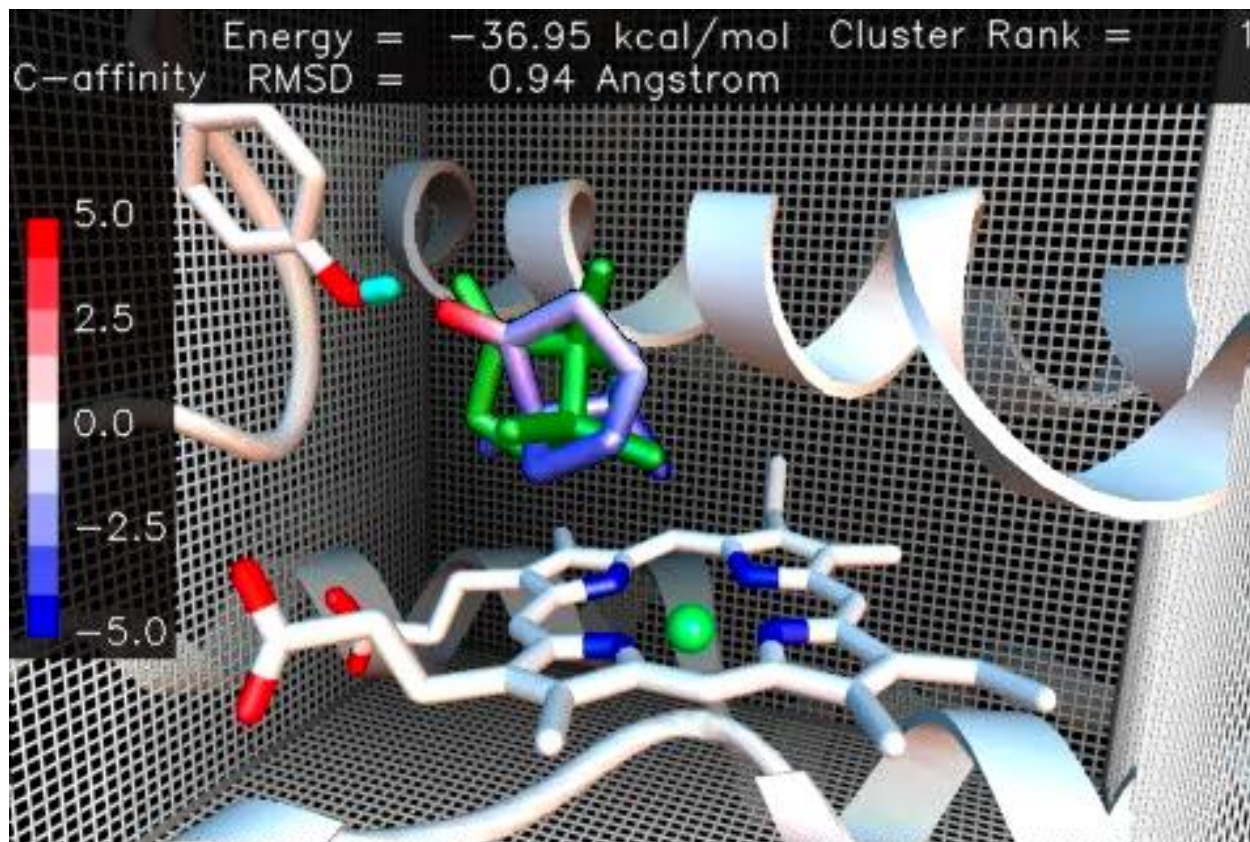
- A molécula de receptor é considerada rígida, apenas o ligando é flexível
- Quanto mais fina for a grelha, mais detalhado será o cálculo (e mais lento)
- Tem que ser calculada uma grelha para cada tipo de átomo do ligando (tipicamente C, N, O, H,...)

- Pré-cálculo energias de interacção com todos os átomos do receptor de cada um dos tipos de átomos do ligando nos nodos de uma grelha
- Para calcular a energia de uma dada conformação do ligando faz-se simplesmente interpolação dos valores calculadas na grelha. Isto é muito mais rápido do que calcular directamente a energia da cada conformação

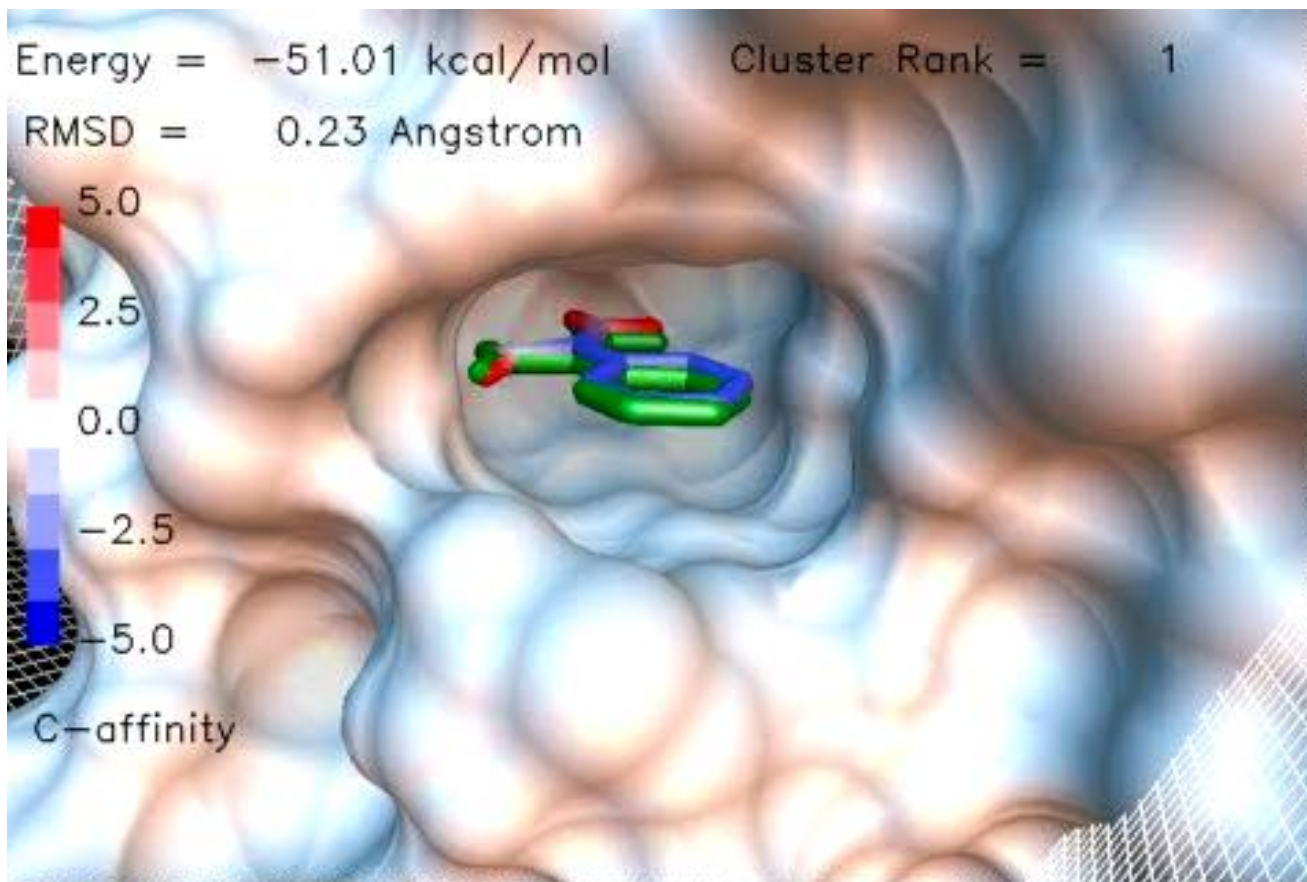
Ligação da biotina à streptavidina



Ligação da cânfora ao citocromo P450cam

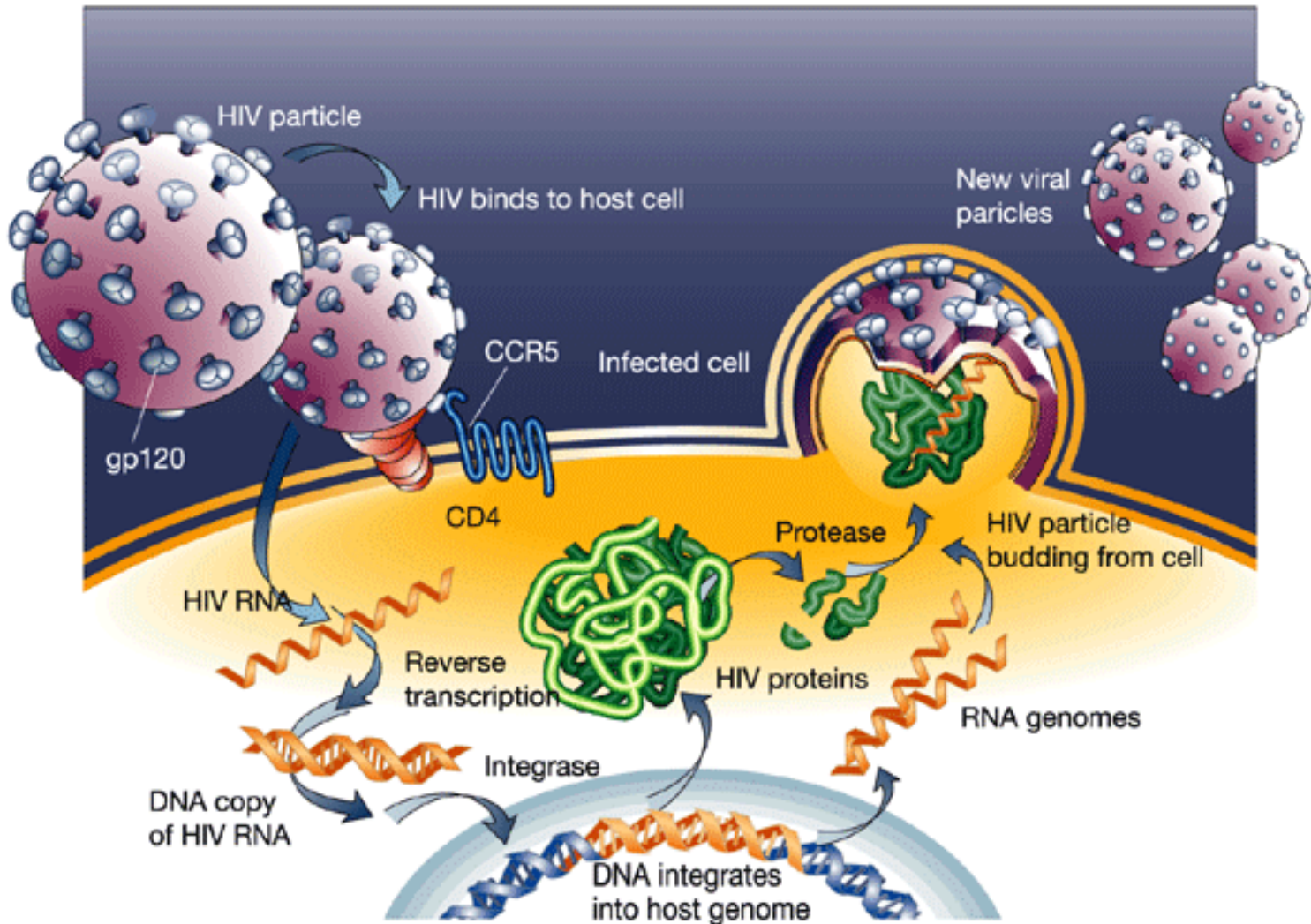


Ligação do inibidor benzamidina à tripsina

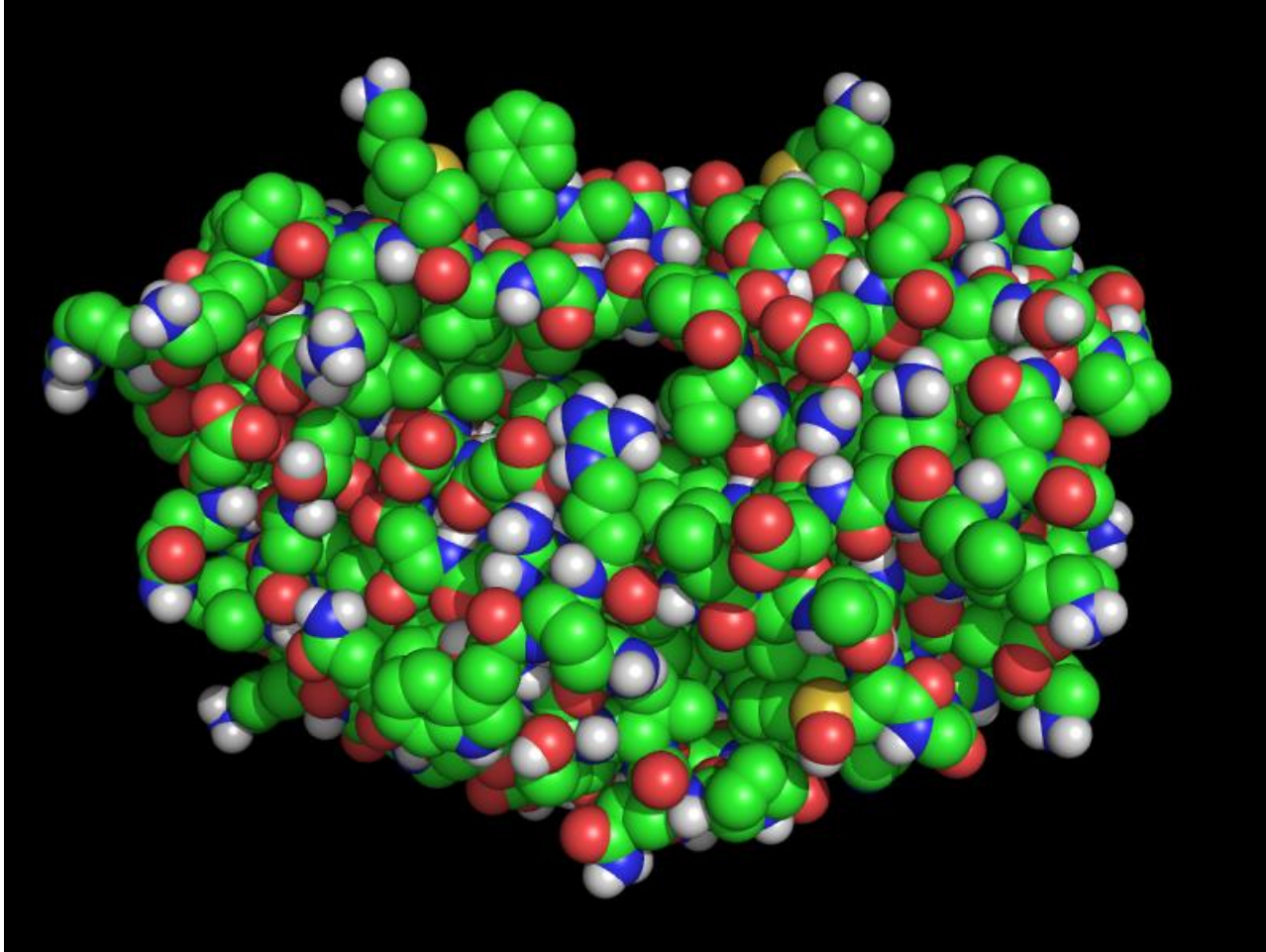


*"Docking" de um inibidor na
protease do HIV*

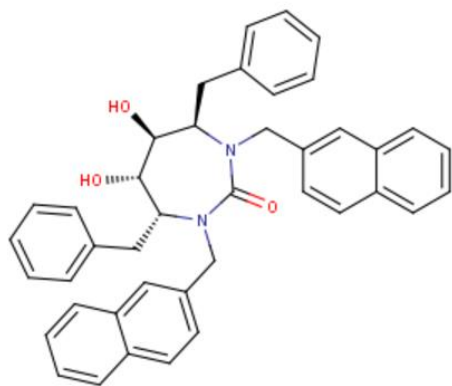
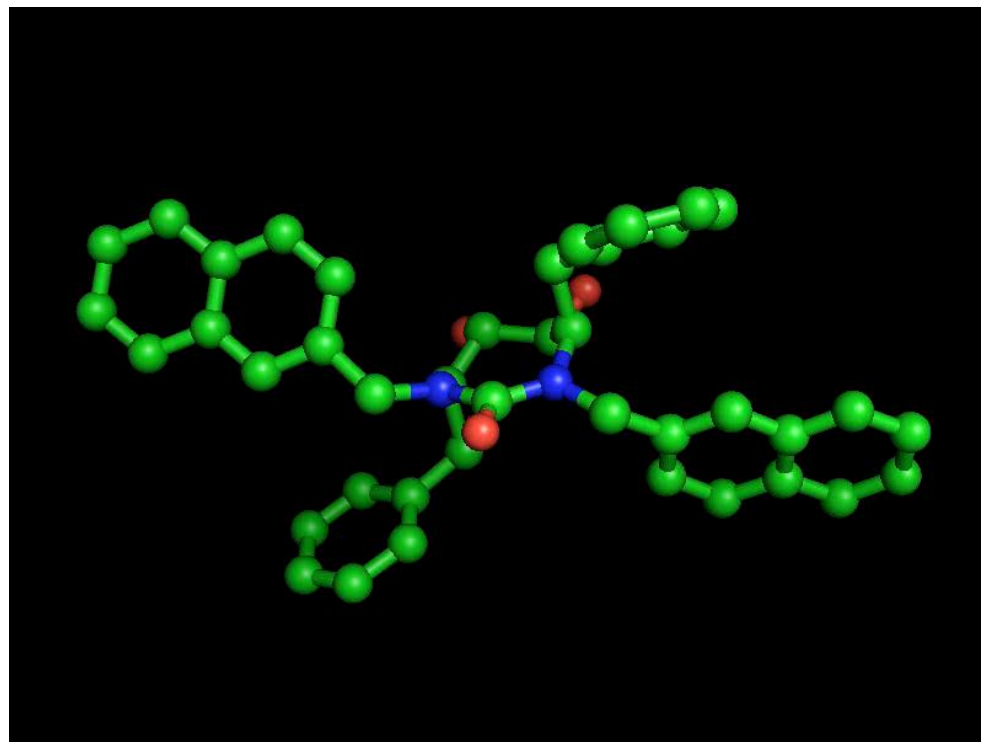
Ciclo de vida do HIV



Protease do HIV-1



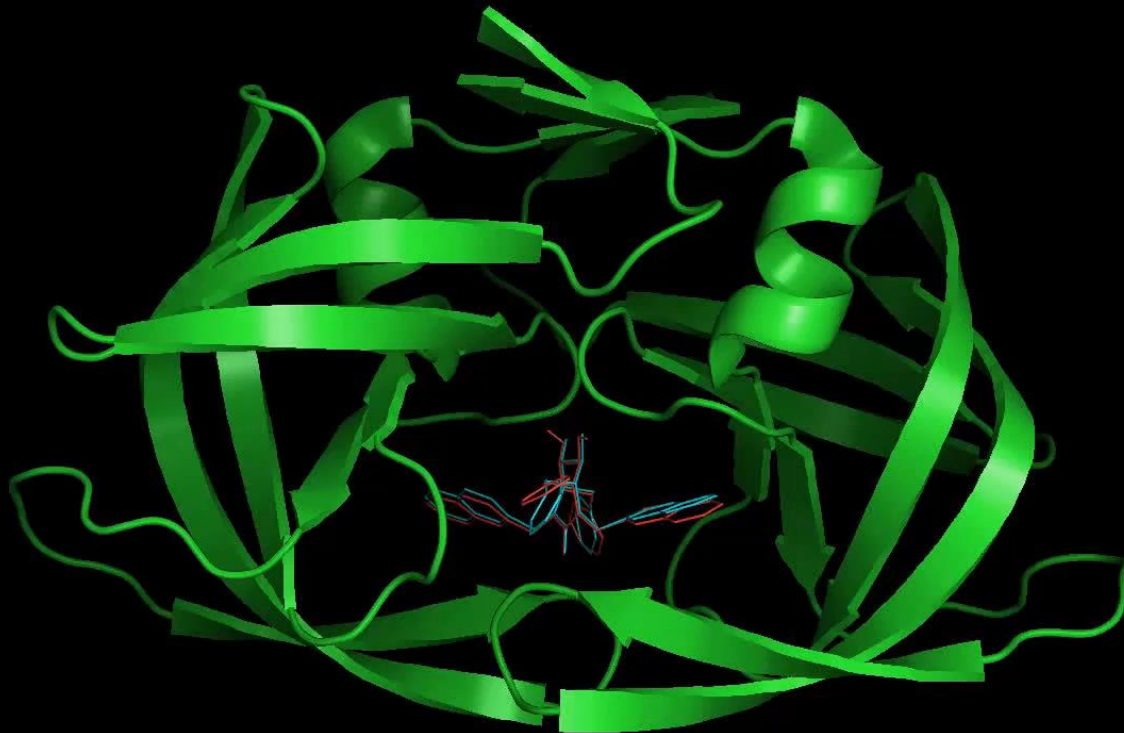
Candidato a inibidor



O xk263 foi um de muitos compostos modelizados, sintetizados e testados para a sua capacidade de inibir a protease do HIV.

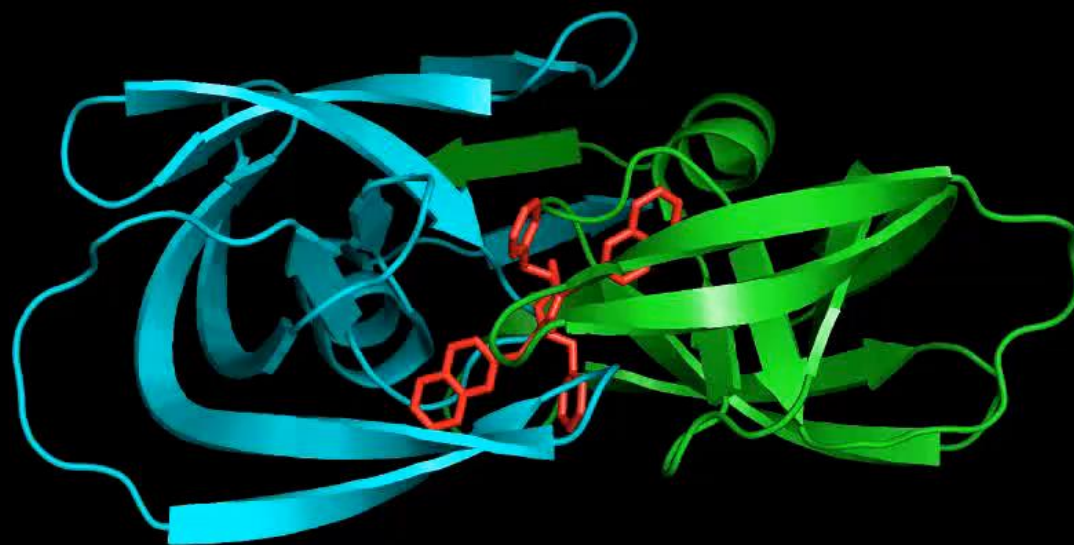
Energias de diferentes soluções obtidas com o software Autodock Vina

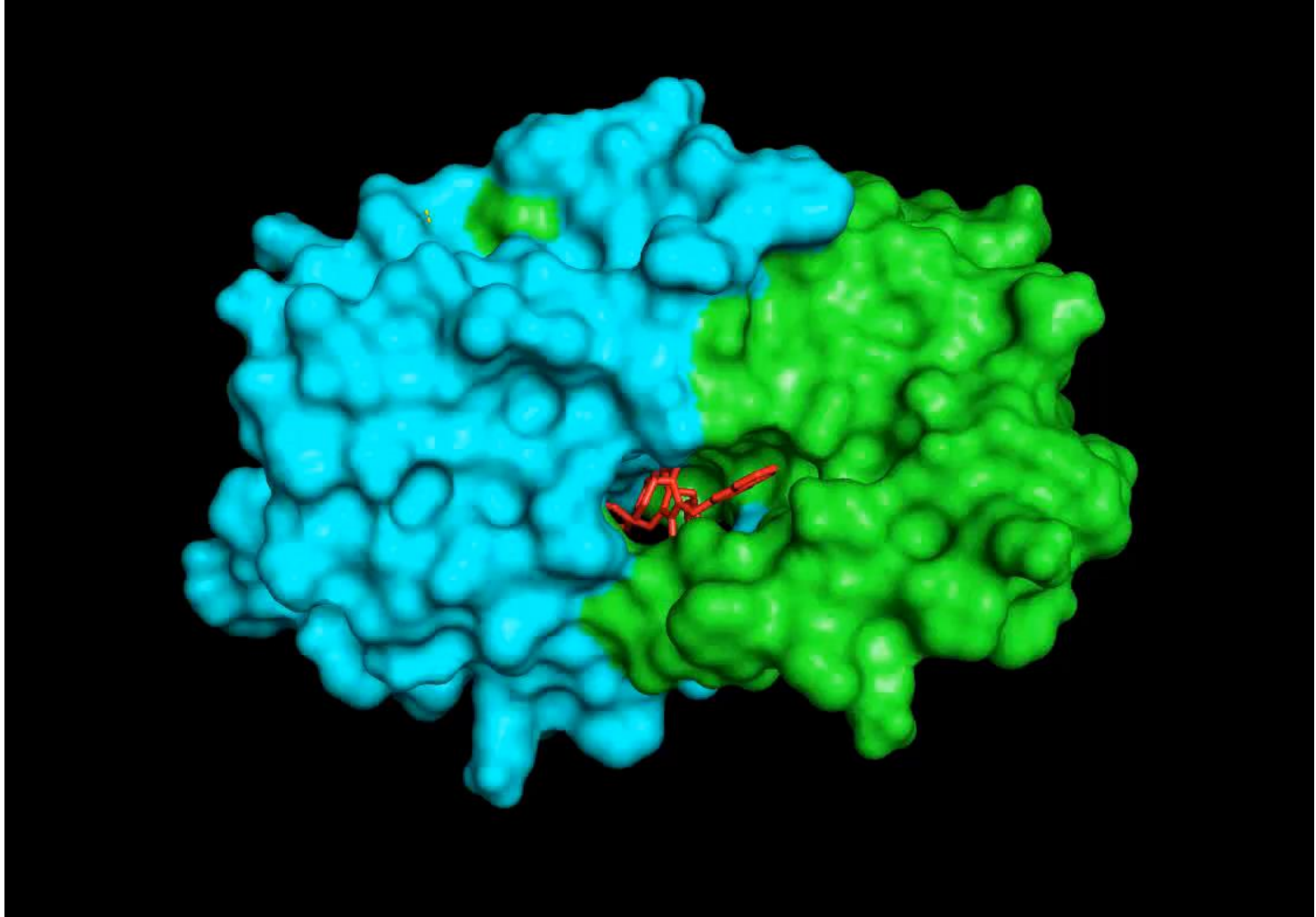
-14.6



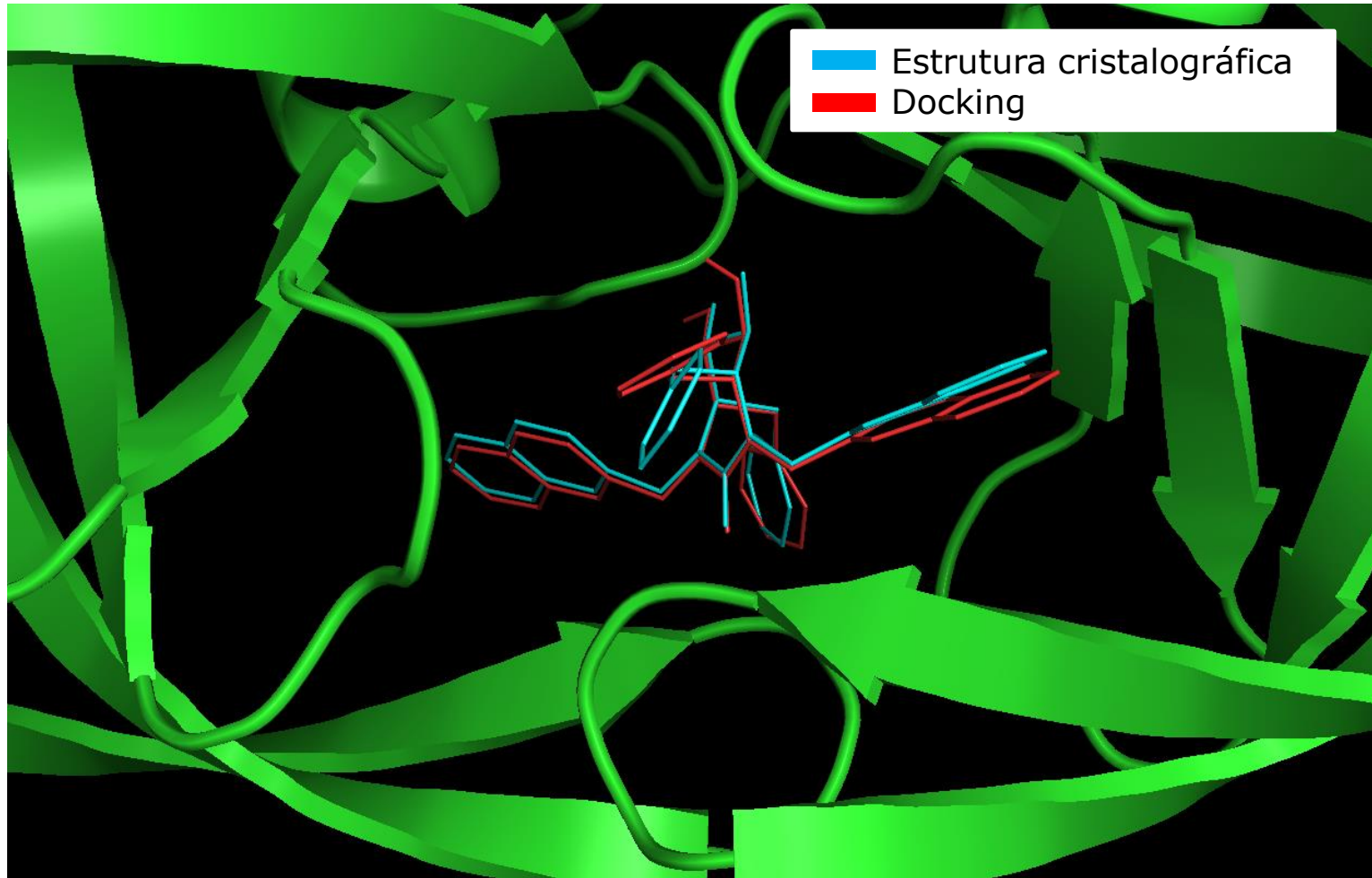
Docking do xk263 no centro activo da protease do HIV







Comparação entre a estrutura cristalográfica e docking com Autodock Vina



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Rational Design of Potent, Bioavailable, Nonpeptide Cyclic Ureas as HIV Protease Inhibitors

Patrick Y. S. Lam,* Prabhakar K. Jadhav, Charles J. Eyermann, C. Nicholas Hodge, Yu Ru, Lee T. Bacheler, James L. Meek, Michael J. Otto, Marlene M. Rayner, Y. Nancy Wong, Chong-Hwan Chang, Patricia C. Weber, David A. Jackson, Thomas R. Sharpe, Susan Erickson-Viitanen*

Mechanistic information and structure-based design methods have been used to design a series of nonpeptide cyclic ureas that are potent inhibitors of human immunodeficiency virus (HIV) protease and HIV replication. A fundamental feature of these inhibitors is the cyclic urea carbonyl oxygen that mimics the hydrogen-bonding features of a key structural water molecule. The success of the design in both displacing and mimicking the structural water molecule was confirmed by x-ray crystallographic studies. Highly selective, preorganized inhibitors with relatively low molecular weight and high oral bioavailability were synthesized.

Knowledge of the HIV protease (HIV PR) mechanism of action and substrate specificity has been extensively used to design a variety of transition state-based inhibitors with inhibition constants in the nanomolar or subnanomolar range (1, 2). The symmetry of the HIV PR dimer guided the design of twofold (C2) symmetric and pseudosymmetric inhibitors (3). However, these inhibitors retain substantial peptide character, and despite many elegant structure-activity studies, it has been difficult to combine adequate potency with oral bioavailability (3, 4). The difficulty in developing such leads into useful therapeutics is challenging, for in addition to the traditional barriers encountered in the drug development process, peptide-based molecules are in general biologically unstable, poorly absorbed, and rapidly metabolized (5). This challenge is not unique to HIV PR; transition from peptide-based leads to therapeutics has proven formidable for other

enzymes such as renin inhibitors (6, 7).

We have previously explored a series of potent, linear C2-symmetric inhibitors in which the transition state mimetic was a diol (8). We were unable to overcome the poor oral bioavailability of these peptide molecules and consequently sought other approaches. The technique of searching databases containing three-dimensional (3D) molecular structures has been used to identify synthetic frameworks that can serve as the starting point for the design of nonpeptide inhibitors, and this approach has been explored with HIV PR. Unfortunately, the HIV PR inhibitors designed to date on the basis of 3D database searches (9, 10) have yielded inhibitors with only micromolar potency.

Our current design of nonpeptide inhibitors (11) began with structural information available from published x-ray crystal structures of HIV-PR inhibitor complexes (12-15). A common feature observed is the presence of a tetracoordinated structural water molecule linking the bound inhibitor to the flexible glycine-rich β strands or "flaps" of the HIV PR dimer (Fig. 1). This water molecule accepts two hydrogen bonds

from backbone amide hydrogens of HIV PR residues Ile 50 and Ile 50' and donates two hydrogen bonds to carbonyl oxygens of the inhibitor, thus inducing the fit of the flaps over the inhibitor (16). Its relevance to the generation of HIV PR inhibitors has been noted (12, 13).

We hypothesized that incorporation of the binding features of this structural water molecule into an inhibitor would be beneficial because its displacement should be energetically favorable (17). In addition, conversion of a flexible, linear inhibitor into a rigid, cyclic structure with restricted conformations should provide a positive entropic effect. Finally, incorporation of a mimic for the structural water within the inhibitor should ensure specificity for the HIV PR as against other aspartic acid proteases, because this water molecule is unique to retroviral proteases. We reasoned that these effects might provide highly potent and specific binding and reduce the need for multiple interactions at the specificity pockets. This should permit design of smaller (<600 daltons) inhibitors with improved oral bioavailability.

Extensive structure-activity relations (SARs) established for C2-symmetric diols indicated that the diol imparts significant potency as compared with corresponding mono-ol transition state analogs (3, 8). Thus, we wanted to incorporate this feature of the diol-HIV PR interaction into a 3D pharmacophore model. However, no x-ray structure of a C2-symmetric diol-protease complex was available when this work was initiated; two independent reports of structures have since appeared (18). Therefore, computer models for C2-symmetric diols bound to the active site of HIV-1 PR were developed from the crystal structure coordinates of a hydroxyethylene inhibitor bound to HIV PR (15) by means of distance geometry (19) (Fig. 2, A and B) and several pharmacophores were generated.

The simplest pharmacophore model (Fig. 2C) was based on two key intramolecular distances: that between symmetric hydrophobic groups, designated P1 and P1', that occupy corresponding enzyme pockets S1 and S1' and that from P1 and P1' to a hydrogen bond donor/acceptor group (or groups) that binds to catalytic aspartates. A 3D database search (20, 21) with this pharmacophore model yielded the "hit" (22) shown in Fig. 2D, which not only met the initial search criteria, but also included an oxygen that matched the structural water found in HIV PR-inhibitor complexes. This 3D search indicated that a phenyl ring could properly position groups to interact with aspartates 25 and 25' as well as to mimic the structural water (Fig. 2E). However, because a phenyl ring might not properly position all substituents in the

inhibitor, a cyclohexanone ring (Fig. 2F) was chosen as the initial synthetic scaffold, with the ketonic oxygen as the structural water mimic. The cyclohexanone ring was enlarged to a seven-membered ring (Fig. 2G) to incorporate the diol functionality. This synthetic target was further modified to a cyclic urea (Fig. 2H) on the basis of two considerations. First, cyclic ureas have established a precedent as excellent hydrogen-bond acceptors both in nature (for example, biotin-streptavidin interactions (23)) and in synthetic systems (24). Second, it was realized that the seven-membered cyclic urea was synthetically accessible by cyclization of the precursor (a phenylalanine-derived diamidinol) used in the linear C2-symmetric diol series (8).

Additional modeling studies were performed with the cyclic ureas to predict the optimal stereochemistry and conformation needed for complementary interaction with the HIV PR. The predicted optimal stereochemistry for cyclic ureas with substituents on the nitrogens is 4R, 5S, 6S, 7R (Fig. 2I), which is derived from unnatural (D) phenylalanine. This is in contrast to the linear C2-symmetric diol inhibitors where natural (L) phenylalanine at P1/P1' provides optimal stereochemistry (25). The optimal stereochemistry prefers the conformation with pseudo diaxial benzyl groups and pseudo diequatorial hydroxyl groups (26). On the other hand, if the nitrogens of the cyclic ureas are not substituted, enantiomers derived from L- or D-phenylalanine fit equally well at the active site. These conformational preferences were subsequently confirmed by small molecule x-ray crystallography (27). Thus, the N-substituted cyclic urea ring provides a rigid framework such that with the proper stereochemistry and conformation, the P1/P1'/P2/P2' and hydroxyls should be highly complementary to the corresponding S1/S1'/S2/S2' pockets and catalytic aspartates of HIV PR.

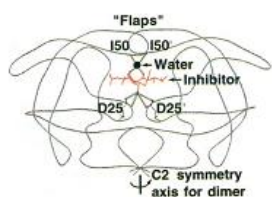


Fig. 1. Key features of the HIV PR dimer. HIV PR is a C2-symmetric dimer, with paired aspartic acid residues located at the floor of the active site and a water molecule juxtapositioned between the inhibitor and flexible enzyme flaps. The complex shown is that of Swain *et al.* (15).

While the above design and modeling studies were based on a static, x-ray structural view of the HIV PR active site, the design insights and predictions were confirmed by the biological results and subsequent structural studies. Cyclic ureas derived from L- and D-phenylalanine, when unsubstituted at the urea nitrogens, have essentially identical inhibition constant (K_i) values (3000 nM and 4500 nM, respectively), in agreement with modeling. Nitrogen-substituted analogs of L-phenylalanine-derived cyclic urea do not result in increased potency as compared with the unsubstituted cyclic urea (for example, the N,N-bis-allyl analog has a K_i of 4500 nM).

There are several factors responsible for the potency of nitrogen-substituted cyclic ureas: (i) the cyclic ureas are preorganized (28) for highly complementary binding to HIV PR, with the conformational entropic penalties typically associated with binding a linear, flexible inhibitor being "prepaid" during synthesis rather than during binding; (ii) displacement of the water molecule is probably thermodynamically favorable (17); and (iii) hydrophobic interactions between the nitrogen substituents and the S2 and S2' subsites of HIV PR are optimized with the preferred conformation and stereochemistry.

As part of our discovery program on HIV PR inhibitors, criteria for development of a compound as an oral agent for the treat-

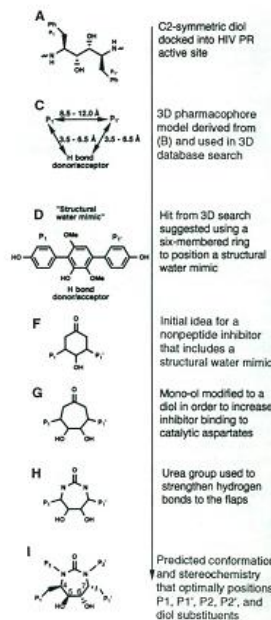
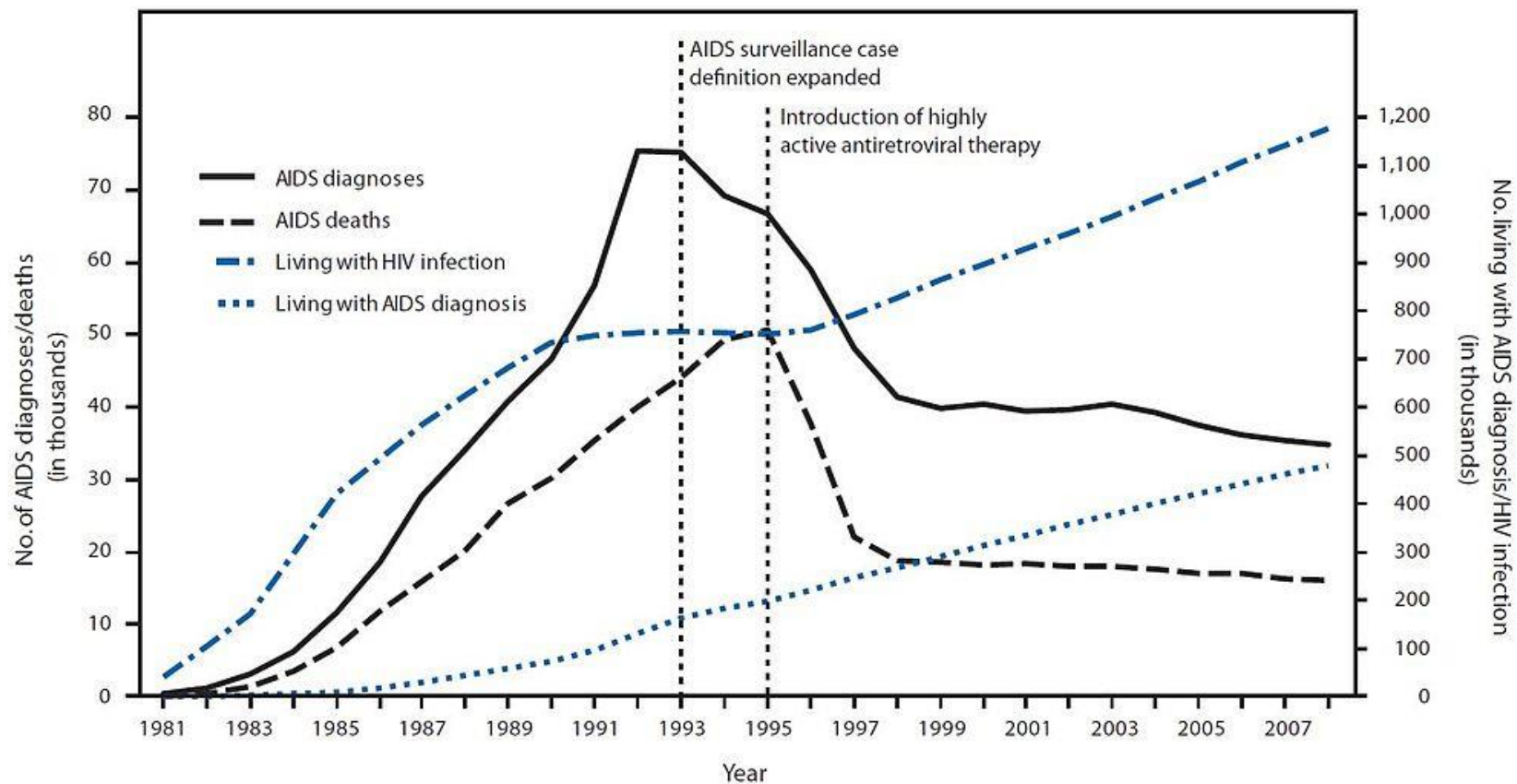
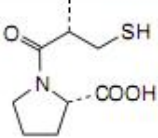
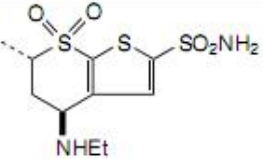
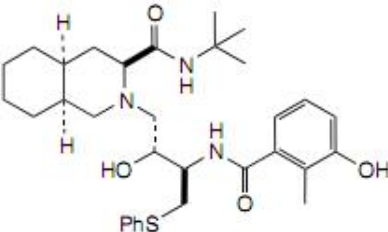
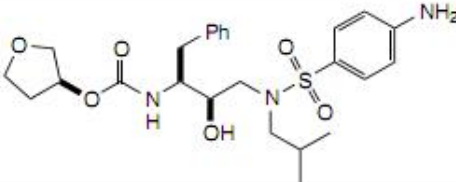
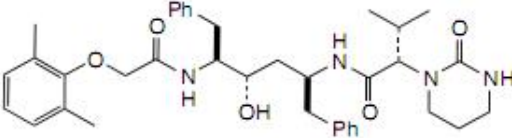
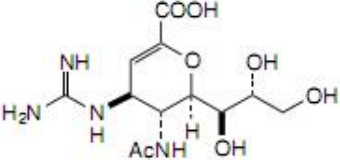
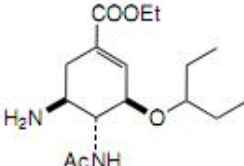


Fig. 2. Strategy and steps involved in the design of cyclic urea inhibitors of HIV PR.

Departments of Virology Research and Chemical and Physical Sciences, The DuPont Merck Pharmaceutical Company, Wilmington, DE 19880.

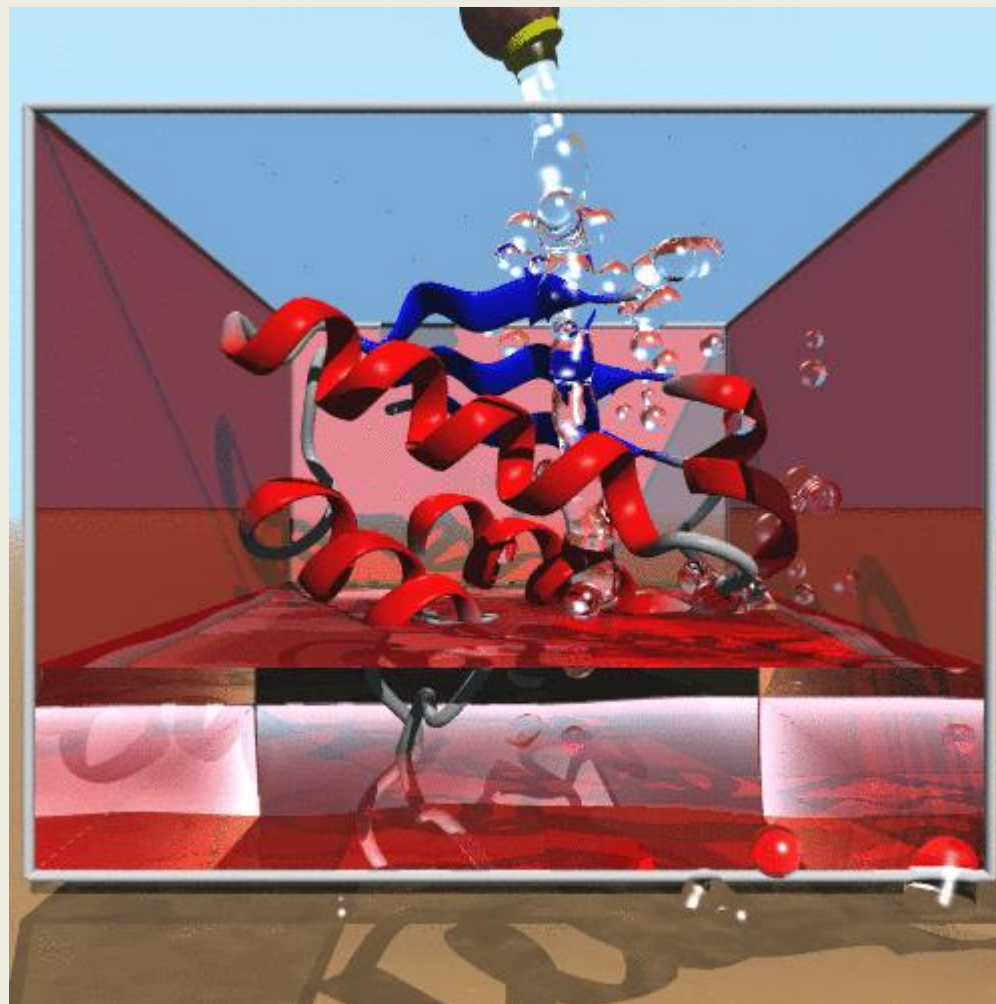
*To whom correspondence should be addressed.



Trade name ³	Generic name	Structure	Mechanism	Disease area	Date launched	Company
Capoten	Captopril		Angiotensin-converting enzyme	Hypertension	1981	Bristol-Myers Squibb
Trusopt	Dorzolamide		Carbonic anhydrase	Glaucoma	1995	Merck
Viracept	Nelfinavir		HIV protease	HIV/AIDS	1999	Agouron (Pfizer) and Lilly
Agenerase	Amprenavir		HIV protease	HIV/AIDS	1999	Vertex and GSK
Aluviran	Lopinavir		HIV protease	HIV/AIDS	2000	Abbott
Relenza	Zanamivir		Neuraminidase	Influenza	1999	Monash University and GlaxoSmithKline
Tamiflu	Oseltamivir		Neuraminidase	Influenza	1999	Gilead and Roche

Antiretrovirais inibidores de protease

Name	Trade name	Company	Patent	Notes
Saquinavir	Fortovase, Invirase	Hoffmann–La Roche	U.S. Patent 5,196,438  	It was the first protease inhibitor approved by the FDA (December 6, 1995).
Ritonavir	Norvir	Abbott Laboratories	U.S. Patent 5,541,206  	-
Indinavir	Crixivan	Merck & Co.	U.S. Patent 5,413,999  	-
Nelfinavir	Viracept	Japan Tobacco	U.S. Patent 5,484,926  	-
Amprenavir	Agenerase	GlaxoSmithKline	U.S. Patent 5,585,397  	The FDA approved it April 15, 1999, making it the sixteenth FDA-approved antiretroviral. It was the first protease inhibitor approved for twice-a-day dosing instead of needing to be taken every eight hours. The convenient dosing came at a price, as the dose required is 1,200mg, delivered in eight very large gel capsules. Production was discontinued by the manufacturer December 31, 2004, as it has been superseded by fosamprenavir.
Lopinavir	Kaletra	Abbott	-	Is only marketed as a combination, with ritonavir.
Atazanavir	Reyataz	-	-	-
Fosamprenavir	Lexiva	GlaxoSmithKline	-	Is a pro-drug of amprenavir. The FDA approved it October 20, 2003. The human body metabolizes fosamprenavir in order to form amprenavir, which is the active ingredient. That metabolism increases the duration that amprenavir is available, making fosamprenavir a slow-release version of amprenavir and thus reduces the number of pills required versus standard amprenavir.
Tipranavir	Aptivus	Boehringer-Ingelheim	-	Also known as tipranavir disodium
Darunavir	Prezista	Tibotec	-	It was approved by the Food and Drug Administration (FDA) on June 23, 2006. Several ongoing phase III trials are showing a high efficiency for the PREZISTA/rtv combination being superior to the lopinavir/rtv combination for first-line therapy ^[1] . Darunavir is the first drug in a long time that didn't come with a price increase. It leapfrogged two other approved drugs of its type, and is matching the price of a third ^{[2][3][4]} .



Limitações do método de MD

- O sistema moleculares tem um elevado número de partículas
- Não há solução analítica para a resolução do sistema de equações de movimento, tem de ser resolvido por métodos numéricos aproximados.
- A integração faz-se com um *step* Δt muito pequeno, tipicamente 0.0005ps a 0.002ps (1ps=10⁻¹² s)
- 100 ps correspondem a 50000 steps de integração para todas as partículas do sistema!
- Com o *hardware* actual conseguem simular-se intervalos de 1-100ns da dinâmica de uma proteína (>100ns para pequenos péptidos) - o record é 1 μ s (1000ns)

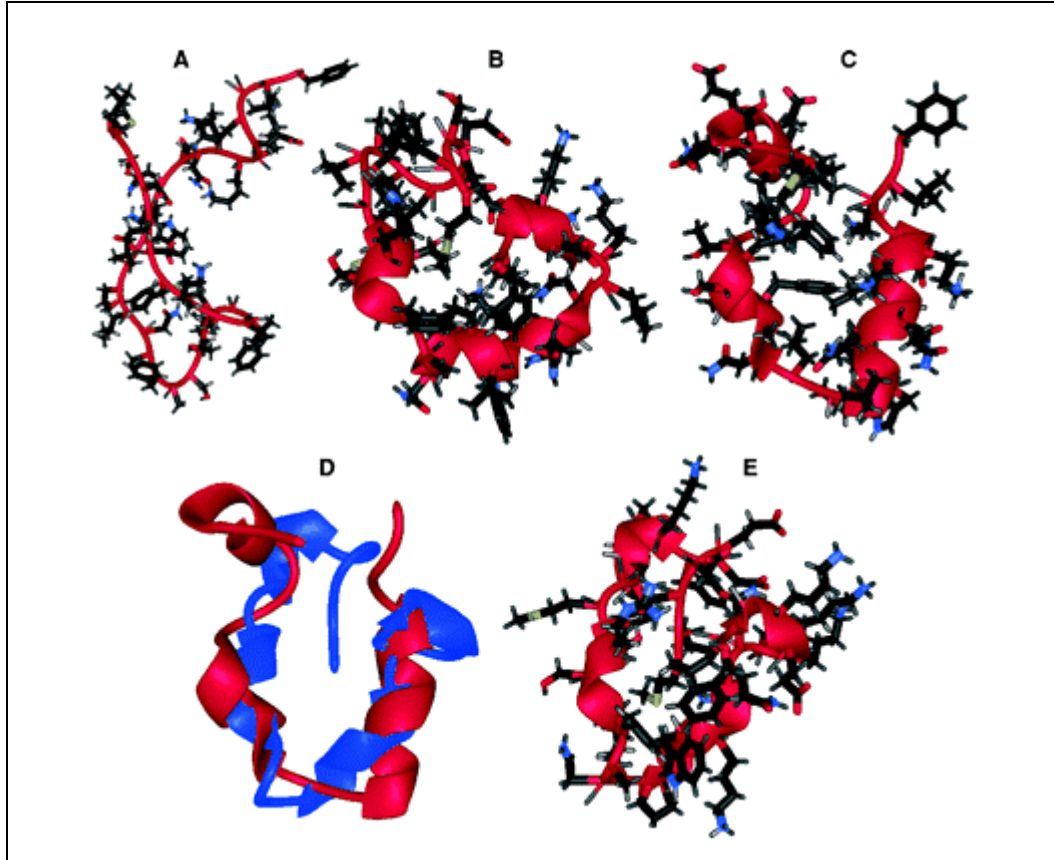
Escala de tempo dos fenómenos estudados por MD:

- | | |
|---|--------------------------|
| • Distensão de ligações | 0.0001 to 0.0001 ns |
| • Rotação de grupos na superfície da proteína | 0.01 to 0.1 ns |
| • Movimento relativo de domínios estruturais | 0.01 to 100 ns |
| • Transições conformacionais, folding | 10000 to 100000000000 ns |



Inacessíveis, mas não
por muito tempo...

State of the art: 1 μ s MD simulation of a small protein



- Villin headpiece: small 36-residue chain that folds spontaneously in about 1-4 μ s
- Simulation box with 3000 water molecules
- Simulation took 4 months CPU time on massively parallel computer (Cray T3)
- A transient state was observed for about 150 ns which resembles the native state

- A:** Initial state
B: Partially folded (980 ns)
C: Native state
D: Native state (red), simulation (blue)
E: transient state